

Efficient Allocations with Moral Hazard and Hidden Borrowing and Lending*

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Abstract

This paper studies the effect of secret access to the credit market in the dynamic moral hazard model. First, in contrast with previous findings, we derive a weak condition under which the efficient allocation improves with respect to the case where the agents can self insure only through borrowing and lending. Second, we provide a tractable recursive framework to study the optimal allocation in this setting. The recursive structure is based on a generalized first order approach, whose validity is verified ex post, using a parsimonious numerical procedure based on the recursive formulation itself. Third, we find that hidden asset accumulation changes dramatically the long run properties of the efficient allocation. The immiseration pattern typical of the standard moral hazard model is replaced by properties qualitatively similar to those of self insurance. The discrepancies with respect to self insurance are important but essentially of quantitative nature.

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1 Introduction

In this paper we study the problem of moral hazard when agents have secret access to the credit market. We analyze the efficient allocation of consumption and effort in an environment where risk averse individuals have random income and can borrow and lend at a given risk-free interest rate. Their asset and consumption decisions are private information. Moreover, each individual can affect future income realizations through his effort decision, which is also non-monitorable. The optimal allocation is derived by solving the problem of a risk-neutral planner who provides incentive compatible insurance contracts based only upon income histories.

A large literature studied optimal long-term insurance contracts under moral hazard assuming that agents cannot borrow or save.¹ Rogerson (1985a) shows that preventing the agent from entering the asset market is critical - even in the presence of liquidity constraints - since in the optimal contract, the agent is actually willing to save. It can easily be shown that when the planner can perfectly control the agents' asset holdings, and can contractually restrict their acquisition of additional assets and liabilities, the efficient allocation is the same as the one where the agents have no access to the credit market. In many situations however the planner cannot have perfect control over the agents' wealth and consumption. This is true for instance when there are hidden storage or investment opportunities. In transitional and developing countries, agents often use foreign currency, gold or some other forms of storage of value for self insurance. These forms of asset accumulation are typically not observable by the government. There are also cases where agents can have secret access to domestic or foreign accounts and credit lines. Therefore, relaxing the assumption of perfect observability and contractability on agents' asset holdings and analyzing the resulting optimal allocation is a potentially very valuable exercise from both the theoretical and applied point of view.

Allen (1985) and Cole and Kocherlakota (2001a) (ACK) study the effect of hidden asset accumulation in a hidden information moral hazard model. They find that the constrained efficient allocation does not differ from that in a 'pure bond economy,' i.e., the allocation the agents could obtain by insuring themselves through borrowing and lending, without the planner's provision of additional insurance.² This result seems to undermine the relevance of our study.

¹This literature includes, among many others, Townsend (1982), Rogerson (1985a), Spear and Srivastava (1987), Green (1987), Phelan and Townsend (1991). Atkeson (1991) analyses a moral hazard model with borrowing and lending and default. His model differs crucially from ours since he assumes that asset holdings are observable.

²Cole and Kocherlakota (2001a) extends the result of Allen (1985) for the general multiperiod case where the agent can only save, but not borrow.

We will show however that as long as the agents do not have complete control over publicly observable income levels the constrained efficient allocation *always* differs from (i.e., it is welfare improving with respect to) self insurance. This is a weak condition on the primitives of the problem that is clearly not satisfied by the ACK framework. In this sense, we can say that the self insurance result of ACK arises in a small class of situations. Our result has important implications for consumption theory and optimal institutional design.

Introducing hidden asset accumulation into the dynamic moral hazard model raises important methodological complications as the problem fails to have a recursive structure, at least in the usual sense. Fudenberg *et al.* (1990) provide characterizations of efficient allocations in a wide class of dynamic environments where agents' preferences over continuation contracts are common knowledge after any history. Since the level of wealth typically affects the agents' attitude toward risk, hidden borrowing and lending leads to a violation of the common knowledge of preferences assumption.³ In spite of that, by using a generalized *first order conditions approach*, we are able to formulate the problem within the dynamic programming framework. In order to keep track of the marginal value of wealth, together with the agent's expected discounted lifetime utility, we introduce the agent's marginal utility of consumption as an additional endogenous state variable. Intuitively, the recursive formulation can be obtained because the first order approach allows us to write the problem in terms of equilibrium values alone. Then, incentive compatibility guarantees that common knowledge of preferences is maintained along the equilibrium path.

Virtually all existing literature justifies the use of the first order approach by showing that - given the optimal contract - the agent's problem is globally concave. For static moral hazard models or for dynamic models with linear taxation without moral hazard this is a perfectly viable procedure.⁴ In Ábrahám and Pavoni (2004b) (AP) we also tackle this problem analytically, in a two period moral hazard framework with hidden savings, and provide sufficient conditions for concavity. Unfortunately, the derivation of (not too restrictive) sufficient conditions for global concavity becomes difficult (perhaps impossible) in a general multi-period setting such as that

³Fernandes and Phelan (2000) and Doepke and Townsend (2004) propose a way to solve recursively an even wider class of problems than that analyzed by Fudenberg *et al.* (1990). Unfortunately, their methods are not viable when hidden actions belong to a continuum as it is the case for saving decisions in our model. Cole and Kocherlakota (2001b) extend the Abreu *et al.* (1990) framework to a wide class of dynamic games. None of their extensions, however, is of any use for us, since our law of motion for bond holdings does not satisfy the 'full support' assumption required there.

⁴See, e.g., Rogerson (1985b) and Jewitt (1988) for the analytic first-order approach in the static moral hazard model; and Chang (1998) and Phelan and Stacchetti (2001) for a similar analytic approach in optimal linear taxation models without moral hazard.

analyzed here.⁵ In this paper, we use a new method, which can be easily extended to a richer class of models with incentive constraints. First, using the first order conditions approach, we solve the (relaxed) problem recursively. Then we take advantage of the dynamic programming formulation and develop a numerical procedure to *verify ex post* whether the obtained allocation is in fact incentive compatible. We allow the agent to re-maximize his lifetime utility taking the optimal (relaxed) transfer scheme as given. Then we check whether the optimal value of the re-maximization problem coincides with that implied by the optimal contract.

This algorithm provides necessary and sufficient conditions for the validity of the first order approach.⁶ Moreover, the procedure turns out to be numerically parsimonious since the agent's re-maximization exercise requires the solution of a simple dynamic programming problem with only one endogenous state. We also design an artificial testing environment and study the effectiveness of the ex post verification procedure in the presence of approximation errors. To explore the distribution of the proposed test statistic across different models we use a battery of two period examples where the agent problem is either globally concave or 'close' to be concave.

The introduction of hidden assets changes dramatically the time series properties of the efficient allocation. It is well-known that in a repeated moral hazard model without hidden asset accumulation, the optimal contract provides a consumption pattern which implies a decreasing lifetime utility for the agent.⁷ This property will not hold in our model. The efficient allocation, adjusted by the presence of the agents' saving incentives, displays (on average) increasing consumption and lifetime utility, key properties of the bond economy. The qualitative properties of the time series of effort (production) and asset holdings also change a lot.

The welfare in the efficient allocation is considerably higher than in the case of self insurance.⁸ Such an improvement is essentially due to quantitative adjustments that do not imply important qualitative changes in the main intertemporal characteristics of the efficient allocation of consumption and effort. One important dimension of welfare improvement is consumption smoothing. In

⁵Williams (2003) proposes an interesting and tractable model in continuous time. However, the conditions for concavity based on the Hamiltonian obtained there are not satisfied in a context where there is a linear intertemporal transfer technology such as that assumed here for savings.

⁶In appendix D we will see that there are interesting cases where the agent's problem is not globally concave but the use of the first order approach is justified.

⁷The most extreme manifestation of this phenomenon is the 'immiserization result' of Thomas and Worral (1990): in a hidden information framework, each agent's lifetime utility goes to minus infinity with probability one in the long run.

⁸We compute the welfare gains for a set of infinite-horizon examples. In the efficient allocation, agents could enjoy an increase in consumption up to between 5.7 and 12.9 percent in all times and states compared to self insurance, depending on the risk aversion parameter.

particular, by decreasing the level of idiosyncratic uncertainty the agent faces, the planner reduces the precautionary motive for savings, and makes the intertemporal path of consumption flatter than that in self insurance. The forces operating in the hidden asset moral hazard economy hence place the average consumption growth patterns *in-between* self insurance and pure moral hazard.

The way we use the first order approach together with the marginal utility of consumption as state variable resembles that adopted in the Ramsey taxation literature by Kydland and Prescott (1980), Chang (1998), and Phelan and Stacchetti (2001). For those models with linear taxation and no moral hazard however the first order approach can easily be justified analytically. One key methodological contribution of this paper is to show an important complementarity between the recursive formulation and the first order approach which allows the formal study of dynamic incentive models where the global concavity of the agent's problem cannot be guaranteed or verified analytically. In an independent work, Werning (2001) develops a similar recursive formulation for the dynamic moral hazard model with hidden savings. Werning (2002) uses it to study optimal unemployment insurance provision. In neither cases he addresses the validity of the first order approach nor investigates the presence and magnitude of welfare gains. Also, the analysis of the optimal unemployment insurance *transfer* scheme in Werning (2002) is essentially orthogonal to our investigation of the behavior of *consumption* and effort.

Because of the methodological problem we mentioned above, the remaining few papers that analyze dynamic moral hazard with non-monitorable asset holdings use particular models and study specific issues. In a two period principal agent relationship, Bizer and DeMarzo (1999) show that hidden access to the credit market reduces total welfare with respect to the no asset market case. They focus on the possibility of increasing welfare by allowing the entrepreneur to default on the debt. We analyze the case where the agent has full commitment, and we show that the welfare reduction highlighted by Bizer and DeMarzo is not as drastic as the ACK analysis would suggest. Bisin and Rampini (2002) study the effect of bankruptcy provision, in a two period model similar to that of Bizer and DeMarzo, where agents have hidden access to insurance contracts and can default on the principal insurer as well. In addition to full commitment, we do not allow agents to secretly trade assets other than a risk free bond. Chiappori et al. (1994) and more recently Park (2004) analyze the optimal contract with discrete effort. They find that under some conditions a renegotiation-proof contract always implements the minimum level of effort. We consider a continuous-effort model, where the planner can commit not to renegotiate the contract ex post. In our framework the optimal allocation of effort is non-degenerate. Kocherlakota (2004) characterizes

the optimal UI transfer scheme in a two-output moral hazard model with hidden savings, where agents' preferences are linear in effort, and effort affects linearly job-finding probabilities. We analyze the optimal contract in a more general model, and we also study the welfare properties of the efficient allocation.

The paper is organized as follows. In the next section, we present our environment, define constrained efficiency, and prove the existence of welfare gains. The recursive formulation and the ex-post verification procedure are presented in Section 3. In Section 4 we study the efficient allocation both analytically and by the mean of an extensive quantitative analysis based on a simple empirical application. Section 5 concludes.

2 Environment and Constrained Efficiency

Environment Consider a small open economy consisting of a large number of agents that are ex-ante identical, and who each live $T \leq \infty$ periods. Each agent is endowed with a private stochastic production technology which takes the following form. There is a finite set $Y = (y^1, \dots, y^N)$ of possible output levels, with $y^i < y^{i+1}$. At each period t , the realization $y_t \in Y$ is publicly observable; however, the probability distribution over Y is affected by the agent's unobservable effort level e , which we assume belongs to a bounded interval $E = [0, e_{\max}]$. The conditional probabilities over Y are defined by the publicly known continuous functions⁹ $p_i(e) = \Pr \{y = y^i \mid e\}$. Hence, agents are subject to idiosyncratic risk, and we assume time independent conditional distribution of income.^{10,11} The history of public outcomes up to period t will be denoted by $h^t = (y_1, \dots, y_t)$.¹²

Agents are allowed to buy and sell a risk-free bond which pays a constant interest rate $r \geq 0$. Their asset holdings are private information, and we assume that each agent is born with no wealth ($b_0 = 0$).¹³ Note that, since agents can trade only risk free bonds, asset markets are incomplete. Therefore, we can expect that a social planner could increase overall welfare by providing additional

⁹We assume continuity of the vector function $\mathbf{p} : E \rightarrow \Delta^N$, where $\Delta^N = \{x \in \mathbb{R}^N \mid x \geq 0, \sum_i x_i = 1\}$.

¹⁰For the variable y we will use, interchangeably, the terms output and income. In the first interpretation, we stress the fact that agents have access to a stochastic production technology. Viewing y as income, emphasizes more that agents are facing idiosyncratic risk (with an endogenous distribution).

¹¹Notice, that this model naturally allows for persistence in idiosyncratic shocks.

¹²Since the only other publicly observable variable is the planner's transfer, and the planner has full commitment, without loss of generality we can restrict public histories to be histories of income realizations alone (see Pavoni, 1999, for details).

¹³In the numerical section, we will also consider $b_0 > 0$. Note that if the initial distribution of assets were not degenerate, in period zero, the planner would propose a menu of long term contracts in order to screen agents with different b_0 's. We do not consider this possibility here, however, since both agents' saving decisions and the interest rate are deterministic, there will be no role for revelation games about wealth after the first period.

insurance. In this setting, a constrained efficient allocation can be computed by solving the problem of a benevolent planner whose aim is to reallocate resources optimally in order to insure agents, subject to the feasibility and incentive constraints which will be specified below.

An *allocation* (or social contract) in this economy is a contingent plan

$$\mathcal{W} = (\boldsymbol{\tau}, \boldsymbol{\sigma}); \quad \boldsymbol{\tau} = \{\tau_t(h^t)\}_{t=1}^T, \quad \boldsymbol{\sigma} = \{\mathbf{e}_t(h^t), \mathbf{b}_t(h^t), \mathbf{c}_t(h^t)\}_{t=1}^T,$$

where $\tau_t(h^t)$ represents the transfer the individual receives in period t , $\mathbf{e}_t(h^t)$ the implemented effort, $\mathbf{b}_t(h^t)$ the bond holdings and $\mathbf{c}_t(h^t)$ the agent's consumption level as a function of the realized history h^t . Note that we assume that agents can only be distinguished through their output histories. In this sense, since all individuals are ex ante identical, we restrict ourselves to symmetric allocations.

To simplify the analysis, we separate the planner's transfer plan, $\boldsymbol{\tau}$, from $\boldsymbol{\sigma}$, the components of the allocation under the agent's control. The metaphor used in contract theory is that the planner proposes $\boldsymbol{\sigma}$, and the plan will be implemented by appropriately designing the transfer scheme $\boldsymbol{\tau}$.

At period t , each agent receives a transfer payment $\tau_t = \tau_t(h^t)$ from the planner, contingent on the realized history h^t . Given today's income level y_t , transfer τ_t , asset level b_{t-1} and the continuation plan $\boldsymbol{\tau} \setminus h^t$, the agent chooses consumption $c_t \geq 0$ and bond holdings b_t subject to the following budget constraint:

$$c_t + b_t = y_t + \tau_t + (1 + r)b_{t-1}. \quad (1)$$

As usual, we impose the condition $b_T \geq 0$ on asset holdings.¹⁴ At the beginning of each period, the agent also decides the effort level $e_t \in E$, which affects the stochastic output realization $y_{t+1} \in Y$, leading to next period output history $h^{t+1} = (h^t, y_{t+1})$. This sequence of events continues until period T is reached.

Agents have intertemporally additive separable von Neumann-Morgenstern utility function. The continuation plan $\mathcal{W} \setminus h^t = \{\tau_\tau(h^\tau), \mathbf{e}_\tau(h^\tau), \mathbf{b}_\tau(h^\tau), \mathbf{c}_\tau(h^\tau) \setminus h^t\}_{\tau=t}^T$ from node h^t generates

¹⁴The equivalent condition for the infinite horizon version of the model ($T=\infty$) is $\lim_{T \rightarrow \infty} \left(\frac{1}{1+r}\right)^T \mathbf{b}_T(h^T) \geq 0$ almost surely for all h^T .

the following expected discounted utility at time $t \geq 1$:

$$\mathbf{U}_t^T(\mathcal{W}; h^t) = \mathbf{U}_t^T(\boldsymbol{\tau}, \boldsymbol{\sigma}; h^t) = \mathbf{E} \left[\sum_{\tau=t}^T \beta^{\tau-t} u(\mathbf{c}_\tau(h^\tau), \mathbf{e}_\tau(h^\tau)) \mid \alpha \setminus h^t \right],$$

where $\alpha \setminus h^t = \{\mathbf{e}_\tau(h^\tau) \setminus h^t\}_{\tau=t}^T$ denotes the implemented effort plan from history h^t onward, $\mathbf{E}$ is the usual expectation operator, and $\beta \in (0, 1)$ is the discount factor. We assume that the choice of $\mathcal{W}$ is implicitly restricted in such a way that both the expectation and the (possibly infinite) summation are well defined. We also assume that the planner always prescribes effort levels which generate full support.¹⁵ We also assume u to be real valued and continuous; strictly increasing, strictly concave in c and decreasing in e .

To be feasible, an allocation $\mathcal{W}$ must be deviation-proof in all components of $\boldsymbol{\sigma}$, that is, in effort e , bond holdings b , and consumption c . Hence, we say that the allocation $\mathcal{W}$ is *sequentially incentive compatible* if, for any history h^t , we have

$$\mathbf{U}_t^T(\boldsymbol{\tau}, \boldsymbol{\sigma}; h^t) \geq \mathbf{U}_t^T(\boldsymbol{\tau}, \tilde{\boldsymbol{\sigma}}; h^t), \quad (2)$$

where $\tilde{\boldsymbol{\sigma}} \setminus h^t = \{\tilde{\mathbf{e}}_\tau(h^\tau), \tilde{\mathbf{b}}_\tau(h^\tau), \tilde{\mathbf{c}}_\tau(h^\tau) \setminus h^t\}_{\tau=t}^T$ is any other possible continuation plan of actions after node h^t , such that $\tilde{\mathbf{c}}_\tau(h^\tau) \geq 0$, $\tilde{\mathbf{e}}_\tau(h^\tau) \in E$, and that given the transfer plan $\boldsymbol{\tau} \setminus h^t$ the budget constraint (1) is satisfied, and either $\tilde{\mathbf{b}}_T(h^T) \geq 0$ or (when $T = \infty$) the appropriate No Ponzi Game condition is satisfied. We denote *the set of incentive feasible allocations* as

$$\Omega = \left\{ \mathcal{W} : \text{for all } h^t \text{ satisfies } \mathbf{c}_t(h^t) \geq 0, \mathbf{e}_t(h^t) \in E, (1), (2), \text{ and } \mathbf{b}_T(h^T) \geq 0 \right\}.$$

Constrained Efficiency For technical tractability, we define the optimal contract as the one that maximizes the planner's net returns (or minimizes costs), subject to incentive feasibility and to the social restriction that each individual must receive at least an expected discounted utility level U_0 . We will then choose U_0 so that the planner's expected discounted returns equal zero. The planner is represented by a risk neutral principal who faces the same interest rate as the agents,

¹⁵This last condition is a weak version of the *full support* assumption typically adopted in the moral hazard literature:

$$p_i(e) > 0 \text{ for any } i = 1, 2, \dots, N \text{ and } \forall e \in E,$$

which requires full support also for actions which might be never implemented in the optimal contract. The reason we keep it in this weaker form will become clear below when we discuss Proposition 1 and its relationship with ACK.

whose net return at node h^t induced by the continuation plan $\mathcal{W} \setminus h^t$ is

$$\Pi_t^T(\mathcal{W}; h^t) = \mathbf{E} \left[\sum_{\tau=t}^T \frac{(1+r)^t}{(1+r)^\tau} (-\tau_\tau(h^\tau)) \mid \alpha \setminus h^t \right].$$

Given the social restriction U_0 , the T -horizon planner's problem can then be formulated as follows:

$$\sup_{\mathcal{W} \in \Omega} \Pi_0^T(\mathcal{W}); \text{ s.t. } \mathbf{U}_0^T(\mathcal{W}) \geq U_0, \quad (3)$$

where

$$\Pi_0^T(\mathcal{W}) = \sum_{i=1}^N p_i^0 \Pi_1^T(\mathcal{W}; y^i), \text{ and } \mathbf{U}_0^T(\mathcal{W}) = \sum_{i=1}^N p_i^0 \mathbf{U}_1^T(\mathcal{W}; y^i)$$

for some initial distribution $\mathbf{p}^0$. We postpone the issue of existence until Section 3.

Under standard assumptions, the allocation in our model is Pareto inferior compared to the case where the planner is able to observe the consumption-saving decision of the agent as the allocation under observable assets is not incentive compatible (e.g., Bizer and DeMarzo, 1999). We mentioned above that Allen and Cole and Kocherlakota (ACK) show that the possibility of secret borrowing and lending may make this efficiency loss extreme. In their case, the planner has no role in enhancing welfare as the only incentive compatible allocation will provide no additional insurance to the agents compared to a pure bond economy. The following proposition shows that in our framework this is typically not the case.

Proposition 1 *Assume $T < \infty$, existence, and that u is either unbounded below or unbounded above or both. Moreover, assume that there exist a subset $I \subset Y$ of output levels containing at least two elements (i.e., $\#I \geq 2$), for which: if $y^i \in I$ then $p_i(e) > 0 \forall e \in E$. Then self-insurance cannot solve (3), i.e. it cannot be a constrained optimal allocation.*

The intuitive idea of this result is simple (the proof are all reported in Appendix A). The planner can *always* provide additional insurance with respect to self insurance, at least in the last two periods of the program. To provide risk sharing the planner makes positive transfers when the publicly observable income level y is low and levies taxes when y is high. Of course, this scheme is implementable only if the agent's control on income levels is imperfect so that he cannot always avoid paying taxes. In Appendix B we provide a counter-example where the assumption $\#I \geq 2$ is not met and we get back the self insurance result.

Notice that Proposition 1 does not only imply that examples where the efficient allocation

improves on self insurance are typical in our framework. It guarantees that as long as $\#I \geq 2$ no matter how one ‘twists’ the technology $(\mathbf{p}, Y)$ and the preferences u , and how one chooses the initial entitlement U_0 , there is not a single case within the whole class of such models where consumption moves as in self insurance. It turns out that with an innocuous change of timing, our framework formally nests that of ACK.¹⁶ In terms of our model, ACK study a special case where the agent has perfect and costless control over publicly observable income levels (i.e., the assumption $\#I \geq 2$ is violated in that framework).

More importantly, this result suggests that there are potential gains from several policy interventions in environments with inherent informational problems (e.g., public unemployment insurance (UI), disability insurance, retirement, income taxation, liquidity provision), even when the public authority does not have control over individuals’ wealth.¹⁷ Moreover, the presence of welfare gains implies that the constrained efficient allocation can be observationally distinguished from the bond economy one. In this sense, our model extends and complements the literature on consumption theory with endogenous incomplete markets, providing a new testable framework (see, Ábrahám and Pavoni, 2005; and Attanasio and Pavoni, 2005).

3 Recursive Formulation and Ex-post Verification

It should not be difficult to see that condition (2) defines a very complicated and enormous set of constraints: already in the two period version of the model the number of constraints is a bidimensional continuum. Perhaps more importantly, there is no tractable way of writing this problem recursively in its original form. Along the lines of Spear and Srivastava (1987) and Green (1987), Abreu *et al.* (1990) show that when agents’ preferences over continuation contracts are common knowledge after any history, the efficient allocation can be characterized by using a one dimensional state variable: the agent’s continuation utility. Notice that the assumption of common knowledge of preferences is not satisfied in our framework. Essentially, the possibility of hidden asset accumulation introduces an adverse selection problem in each period since agents with different asset

¹⁶This is formally shown in Abraham and Pavoni (2004a).

¹⁷Hopenhayn and Nicolini (1997), Pavoni (2003a,b), Werning (2002) and Kocherlakota (2004) study the design of an optimal UI scheme using a dynamic moral hazard framework, where the worker’s job search effort level is private information. A dynamic moral hazard model with observable savings is used by Diamond and Mirrlees (1978) to analyze social insurance and retirement. Golosov and Tsyvinski (2003) use a similar model to study the design of an optimal disability insurance scheme. Kocherlakota et al. (2003) study optimal income and capital taxation in a dynamic framework with moral hazard à la Mirrlees (1971). Optimal liquidity provision in presence of moral hazard is studied for example in Diamond and Dybvig (1983).

levels respond differently to the contract. Fernandes and Phelan (2000) and Doepke and Townsend (2003) show that this adverse selection problem can be resolved by using one state variable for each agent type. According to this approach, in our framework with a continuum of possible asset levels, the number of types explodes and the relevant state becomes a *function*, i.e. an infinite dimensional object. These infinities pose obvious computational difficulties that make this approach infeasible in practice within our model (for similar discussions see also Phelan and Stacchetti, 2001; and Kocherlakota, 2004).

In this section, we adopt a generalized first order approach which solves both aforementioned problems (the numerosity of the incentive constraints and the tractability of the recursive formulation). Moreover, we take advantage of the dynamic programming framework and develop a numerical procedure to verify ex-post the validity of the first order approach.

3.1 The First Order Conditions Approach

From now onward, we assume that both u and $\mathbf{p}$ are differentiable. The adoption of the first order approach means that the set of constraints described in (2) are replaced by the agent's corresponding first order conditions along the optimal path. Using the budget constraint (1) to eliminate the planner transfers τ_t , and assuming interiority with respect to e ,¹⁸ for any $h^t \neq h^T$ the agent's first order conditions become

$$e : \quad -u'_e(\mathbf{c}_t(h^t), \mathbf{e}_t(h^t)) = \beta \sum_i p'_i(\mathbf{e}_t(h^t)) \mathbf{U}_{t+1}^T(\mathcal{W}; (h^t, y^i)); \quad (4)$$

and

$$b : \quad u'_c(\mathbf{c}_t(h^t), \mathbf{e}_t(h^t)) \geq \beta(1+r) \sum_i p_i(\mathbf{e}_t(h^t)) u'_c(\mathbf{c}_{t+1}(h^t, y^i), \mathbf{e}_{t+1}(h^t, y^i)). \quad (5)$$

Notice, that the asset level b_t does not enter any of these constraints. This is essentially due to fact that the agent faces the same interest rate as the planner does. Now, define the set of social

¹⁸We will never consider the possibility that the upper bound on E is binding. However, we might easily allow for a more general formulation of the following form:

$$u'_e(\mathbf{c}_t(h^t), \mathbf{e}_t(h^t)) + \beta \sum_i p'_i(\mathbf{e}_t(h^t)) \mathbf{U}_{t+1}^T(\mathcal{W}; (h^t, y^i)) \leq 0$$

with equality if $\mathbf{e}_t(h^t) > 0$. Moreover, notice that when $T < \infty$ we must impose the obvious corner solution $\mathbf{e}_T(h^T) = 0$.

contracts satisfying these first order conditions as

$$\Omega_{FOC} = \{ \mathcal{W} : \text{for any } h^t \text{ satisfies } \mathbf{c}_t(h^t) \geq 0, \mathbf{e}_t(h^t) \in E; (1), (4) \text{ and } (5) \}.$$

With some abuse of notation, we will also consider contract continuations $\mathcal{W} \setminus h^s$ belonging to the set Ω_{FOC} . In these cases the restrictions are obviously only related to histories after node h^s .

3.2 The Recursive Problem

In this section we focus on the infinite horizon case, and in order to simplify notation, we do not report the superindex $T = \infty$. The ‘true’ value function of the relaxed problem is thus defined as follows:

$$\begin{aligned} V_{foc}^*(U_0) &= \sup_{\mathcal{W} \in \Omega_{FOC}} \Pi_0(\mathcal{W}) \\ \text{s.t. } & \mathbf{U}_0(\mathcal{W}) \geq U_0. \end{aligned} \tag{6}$$

We argued above that when consumers can secretly borrow and lend, the state variable used by the standard recursive contracting literature - the agent’s lifetime utility U - is no longer sufficient to describe the constrained efficient allocation. We will see that recursivity can be recovered by complementing lifetime utility with an additional endogenous state: the marginal utility of consumption $u'_c(c, e) = x$. From the Euler equation it is clear that by affecting the marginal value of wealth $\beta \mathbf{E}[(1 + r) u'_c(c_{t+1}, e_{t+1})]$ the planner can fully control agents’ asset decisions at the margin. The adoption of $u'_c(c, e)$ as a state variable becomes now natural since in our framework both r and β are exogenously fixed at a constant level.

Our relevant state space is hence two-dimensional.¹⁹ However, before applying the recursive techniques of Stokey *et al.* (1989) (SLP) to our problem, we have to consider the possibility that the feasibility correspondence can be empty for some combinations of the states. To overcome this complication, we could basically follow two alternative procedures. The first possibility is the use of infinite penalizations. This option basically sets the value of the planner to minus infinity for each combination of states (U, x) which cannot be implemented by any incentive feasible

¹⁹Notice, that in our repeated framework the exogenous state y does not affect the space of endogenous states U and x .

contract.²⁰ We choose to follow a second approach, which typically delivers a continuous value function. This second procedure is divided into two main steps. It first derives the state space (or domain restriction) M^* : the set of combinations (U, x) for which a relaxed incentive compatible contract delivering lifetime utility U , and a marginal utility x to the agent in period zero does exist. In the second step of the procedure, the problem is solved using usual recursive techniques.

At each node h^t the (time invariant) set of relaxed incentive feasible endogenous states is formally defined as follows:

$$M^* = \{(U, x) \in \mathfrak{R}^2; \exists \mathcal{W} \setminus h^t \in \Omega_{FOC} ; u'_c(\mathbf{c}_t(h^t), \mathbf{e}_t(h^t)) \leq x, \text{ and } \mathbf{U}_t(\mathcal{W}; h^t) = U\}.$$

It is easy to see that M^* is non empty.²¹ Given M^* we have the following:

Proposition 2 *Given an initial distribution $\mathbf{p}^0$, and an ex-ante expect discounted utility level U_0 guaranteed to the agent, the value function V_{foc}^* solves*

$$\begin{aligned} V_{foc}^*(U_0) &= \sup_{\substack{(x^i, U^i) \in M^* \\ i = 1, \dots, N}} \sum_i p_i^0 V(y^i, U^i, x^i) \\ \text{s.t.} \quad &: \sum_i p_i^0 U^i \geq U_0, \end{aligned} \tag{7}$$

where $V : Y \times M^* \rightarrow \mathfrak{R}$ is a solution to the following functional equation

$$\begin{aligned} V(y, U, x) &= \sup_{\substack{(x^i, U^i) \in M^* \\ e \in E, c \geq 0}} y - c + \frac{1}{1+r} \sum p_i(e) V(y^i, U^i, x^i) \\ \text{s.t.} \end{aligned} \tag{8}$$

²⁰See Rockafellar (1975), and Rustichini (1998).

²¹For example, the full insurance contract with $\mathbf{e}_t(h^t) \equiv 0$ is always feasible. In this case, for any given $x \in \mathfrak{R}_{++}$ the corresponding utility level is obtained by $U = \frac{u(g(x), 0)}{1-\beta}$, where g is the consumption component of the inverse of the marginal utility when $e = 0$.

$$U = u(c, e) + \beta \sum_i p_i(e) U^i \quad (9)$$

$$-u'_e(c, e) = \beta \sum_i p'_i(e) U^i \quad (10)$$

$$u'_c(c, e) \geq \beta(1+r) \sum_i p_i(e) x^i \quad (11)$$

$$x \geq u'_c(c, e). \quad (12)$$

Conversely, if a bounded function V defined on $Y \times M^*$ satisfies the functional equation (8)-(12) and M^* is compact then a solution to (6) exists and

$$V_{foc}^*(U_0) = \max_{(x^i, U^i) \in M^*} \sum_i p_i^0 V(y^i, U^i, x^i), \quad s.t. \quad \sum_i p_i^0 U^i \geq U_0.$$

Notice that V is weakly increasing in x and constant for all values for which constraint (12) does not bind. Moreover, from (11) the choice of low values for x^i is always feasible. Hence x^i can without loss of generality be chosen so that to satisfy (12) with equality.

Let us now briefly turn to the domain restriction M^* . An argument similar to that of Abreu *et al.* (1990) implies that the set M^* can be derived by starting from a sufficiently large set $\overline{M} \supset M^*$, and computing the largest fixed point of the following operator:

$$\mathbf{B}(M) = \{(U, x) \in \mathfrak{R}^2; \exists (U^i, x^i) \in M, e \in E \text{ and } c \geq 0; (9)-(12) \text{ are satisfied}\}. \quad (13)$$

It turns out that $\mathbf{B}$ is monotone²² and maps closed sets into closed sets.²³ Moreover, since the sequence $M_n = \mathbf{B}^n(\overline{M})$ is monotone, it must converge to the set $M_\infty = \lim_{n \rightarrow \infty} \mathbf{B}^n \overline{M} = \cap_{n=1}^\infty M_n$, which is closed as an intersection of closed sets. It can be shown that if $\overline{M}$ is chosen sufficiently large, we have $M_\infty = M^*$ since the sequence converges to the largest fixed point of the operator $\mathbf{B}$. It can be easily shown that when u is bounded with bounded first derivatives, M^* is compact. In the finite horizon version of our model, the domain restriction sets can be similarly computed following a backward procedure.²⁴

²²That is, for any two sets $M, M' \subseteq \mathfrak{R}^2$ if $M \subseteq M'$, then $\mathbf{B}(M) \subseteq \mathbf{B}(M')$.

²³Notice that the constraint set is formed by either equalities or weak inequalities.

²⁴The set M_T for the T -horizon problem can be computed by applying the same map $\mathbf{B}$ we defined above as follows $M_T = \mathbf{B}(M_{T-1}) = \mathbf{B}^{T-1}(M_1)$, where M_1 represents the set of states attainable in a one-period problem:

$$M_1 = \{(x, U) : u'_c(c, 0) \leq x, U = u(c, 0), c \geq 0\}.$$

3.3 Ex-post Verification Procedure of the First Order Approach

Obviously, any interior contract $\mathcal{W}$ that is incentive feasible according to (2) - i.e. $\mathcal{W} \in \Omega$ - is such that $\mathcal{W} \in \Omega_{FOC}$. For the purposes of this paper, we should look at conditions under which an optimal solution $\mathcal{W}^*$ to the following *relaxed* planner's problem:

$$\begin{aligned} \sup_{\mathcal{W} \in \Omega_{FOC}} \quad & \Pi_0(\mathcal{W}) \\ \text{s.t.} \quad & \mathbf{U}_0(\mathcal{W}) \geq U_0, \end{aligned} \tag{14}$$

is such that $\mathcal{W}^* \in \Omega$. In that case, the solution to problem (14) satisfies incentive compatibility, hence we have actually derived the efficient contract.

A direct application of this argument is the basis of our verification of the first order approach. After computing the optimal contract according to (14), the procedure allows the agent to *re-maximize* his lifetime utility by choosing effort, consumption, and bond holdings taking the optimal (relaxed) transfer scheme as given. Then we check whether the optimal decisions of this re-maximization problem coincide with those implied by the relaxed optimal contract, i.e. whether $\mathcal{W}^*$ is actually an incentive compatible allocation.

Notice that for any given transfer scheme $\boldsymbol{\tau} = \{\boldsymbol{\tau}_t(h^t)\}$, the agents' incentive constraint (2) is described by the following (re)maximization problem

$$\begin{aligned} U_0^R(\boldsymbol{\tau}, y) = \sup_{\boldsymbol{\sigma}} \mathbf{E}_0 \left[\sum_{t=1}^{\infty} \beta^{t-1} u(c_t, e_t) \mid \boldsymbol{\alpha} \right] \\ \text{s.t.} \quad \mathbf{c}_t(h^t) + \mathbf{b}_t(h^t) = y_t + \boldsymbol{\tau}_t(h^t) + (1+r)\mathbf{b}_{t-1}; \quad \mathbf{c}_t(h^t) \geq 0, \quad \mathbf{e}_t(h^t) \in E \text{ for any } h^t; \\ \lim_{t \rightarrow \infty} \left(\frac{1}{1+r} \right)^t \mathbf{b}_t(h^t) \geq 0 \text{ a.s.; with } y_1 = y \text{ and } b_0 = 0. \end{aligned} \tag{15}$$

where each consumer chooses contingent plans $\boldsymbol{\sigma}$ of effort, consumption and bond holdings and $\boldsymbol{\alpha}$ is the effort plan implied by $\boldsymbol{\sigma}$. The basic idea of the verification procedure is to allow the agent to solve the above remaximization problem, assuming that the transfer scheme is that implicitly defined in Proposition 2. The first order approach is valid if the agent chooses the same consumption, effort and bond holdings as the one implied by the initial solution.

Proposition 3 *Assume M^* is compact and V bounded. Let $U^*(U_0)$ be the expected discounted*

lifetime utility level obtained by the agent according to (7) with $\mathbf{p}^0$ degenerate at y , and τ be the associated transfer scheme. The first order condition approach is justified if and only if $U_0^R(\tau, y) = U^*(U_0)$.

The proof of Proposition 3 uses the fact that the relevant deviations will always increase the agent's lifetime utility. Since it is always true that $U_0^R(\tau, y) \geq U^*(U_0)$, the task of comparing two allocations is in fact greatly simplified as one only need to compare two real numbers: $U_0^R(\tau, y)$ and $U^*(U_0)$.²⁵ However, since τ is a history dependent stochastic process (i.e. an infinite dimensional object), the verification procedure seems to still require a formidable task. The key advantage of our procedure comes from the observation that the recursive formulation implies that past history can be summarized by the states U_t, x_t .²⁶ In particular, by using the budget constraint and normalizing bond holdings to zero in each period, the disposable income coincides with the policy function for consumption:

$$y_t + \tau_t = c(U_t, x_t).$$

In turn, because of the dynamic programming framework, the states evolve according to the following *time invariant* policy rules:

$$U_{t+1}^i = f(U_t, x_t, y_{t+1}^i) \quad (16)$$

$$x_{t+1}^i = h(U_t, x_t, y_{t+1}^i), \quad i = 1, 2, \dots, N. \quad (17)$$

As a consequence, the re-maximization problem (15) can be written in recursive form as follows:

$$V^R(U, x, b) = \max_{c^R \geq 0, b^R \geq -B(U, x), e^R \in E} u(c^R, e^R) + \beta \sum_i p_i(e^R) V^R(U^i, x^i, b^R) \quad (18)$$

$$\text{s.t.} \quad c^R + b^R \leq c(U, x) + (1 + r)b$$

$$U^i = f(U, x, y^i)$$

$$x^i = h(U, x, y^i),$$

where $-B(U, x)$ is an appropriately defined lower bound for assets to prevent Ponzi schemes.

²⁵Consistently with our recursive approach based on continuation utilities, we disregard payoff-equivalent deviations.

²⁶Since the problem is a repeated one, the relevant policies do not depend on y_t either.

Therefore, given that the policies c , f , and h (i.e. the transfer scheme τ) are exogenous rules for the agent at this stage, the problem to be solved is very similar to that of self insurance, where the only endogenous state variable is the level of bonds b . According to Proposition 3, the first order approach is verified if we have $V^R(U^*(U_0), x^*(U_0), 0) = U^*(U_0)$, where $U^*(U_0), x^*(U_0)$ are the initial values for the states derived in the period zero maximization problem (7), with $\mathbf{p}^0$ degenerate at y .

In Section 4.2.4, we apply the verification procedure to a couple of simple infinite horizon applications, and the procedure seems to justify the use of the first order approach for those cases. Clearly, the presence of approximation errors makes it impossible for $U_0^R(\tau) = U^*(U_0)$ to hold exactly. In Section 4.2.4 and in Appendix D, we propose a normalized ex post deviation measure and explore the ‘statistical’ properties of the proposed test by using a battery of two period examples.

4 Characterizing the Efficient Allocation

In the next section, we explore how the introduction of hidden assets affect the optimal provision of incentives by exploiting the recursive formulation of the model analytically. In Section 4.2, we use a simple empirical application to complement this analysis with a simulation based one mainly focused on observable characteristics such as consumption growth, consumption dispersion and the level of production.

4.1 General Properties of the Optimal Contract

Proposition 2 asserts that, under some conditions, the optimal contract can be fully characterized by analyzing the properties of the value function associated with the maximization problem (8)-(12). It is not easy to derive analytically smooth properties for V however, one can use lotteries on the states and convexify the problem.²⁷ Consider the Lagrangian associated with the *extended*

²⁷The concave function $\mathbf{V}$ would hence be defined as

$$\begin{aligned} \mathbf{V}(U, x, y) &= \max_{(U(z), x(z))_{z \in M^*}} \int_0^1 V(U(z), x(z), y) dz \\ \text{s.t. } U &= \int_0^1 U(z) dz; \quad \text{and} \quad x = \int_0^1 x(z) dz, \end{aligned}$$

static problem (8)-(12) and let's look for a saddle point

$$\begin{aligned}
V(y, U, x) = & \sup_{c \geq 0, e \in E, (x^i, U^i) \in M^*} \inf_{\lambda, \mu, \phi, \xi \geq 0} y - c + \frac{1}{1+r} \sum_i p_i(e) V(y^i, U^i, x^i) + \\
& + \lambda \left[u(c, e) + \beta \sum_i p_i(e) U^i - U \right] + \mu \left[u'_e(c, e) + \beta \sum_i p'_i(e) U^i \right] \\
& + \xi \left[u'_c(c, e) - \beta(1+r) \sum_i p_i(e) x^i \right] + \phi [x - u'_c(c, e)].
\end{aligned}$$

Assuming $\beta(1+r) = 1$, interiority with respect to M^* , and validity of the envelope theorem, we obtain

$$V'_U(y, U, x) = -\lambda \quad (19)$$

$$V'_x(y, U, x) = \phi \geq 0. \quad (20)$$

Our simulations show that in the optimal contract V decreases with U ; hence $\lambda \geq 0$. This is intuitive, since the multiplier λ corresponds to the agent's 'temporary' Pareto weight (à la Marcet and Marimon, 1999). Now consider the first order conditions with respect to U^i :

$$\lambda^i = -V'_U(y^i, U^i, x^i) = \lambda + \mu \frac{p'_i(e)}{p_i(e)}. \quad (21)$$

Condition (21) identifies the optimality properties induced only by the moral hazard part of the model. Because of its informational content on the supplied level of effort, the sign and magnitude of the likelihood ratio $\frac{p'_i(e)}{p_i(e)}$ (and only this) induce proportional variations on the Pareto weight λ^i assigned to the agent.²⁸ Now, take expectations on (21), and use (19) to get

$$V'_U(y, U, x) = \mathbf{E} [V'_U(y^i, U^i, x^i) \mid e]. \quad (22)$$

where z is a payoff irrelevant random variable uniformly distributed on $[0, 1]$; and, similarly to (8), for all $(x, U) \in M^*$ we have

$$\begin{aligned}
V(U, x, y) = & \max_{x^i, U^i, c, e} y - c + \frac{1}{1+r} \sum_i p_i(e) \mathbf{V}(U^i, x^i, y^i) \\
& s.t. \quad (9)-(12);
\end{aligned}$$

and $\partial V(U, x, y) = \partial V(U(z), x(z), y)$ for all $z \in [0, 1]$ where V is superdifferentiable in (U, x) .

²⁸When V is concave, the envelope condition (19) implies that for any given x , a high λ^i is associated to a large level of U^i .

Condition (22) can be interpreted as the planner's Euler equation, where U is the relevant asset, and (9) is the associated budget constraint. According to (22), the planner allocates the agent's lifetime utilities intertemporally in order to maximize his net surplus. While doing so, he must respect the cross-sectional restriction implied by the effort incentive constraint (10). This restriction only allows for reallocations where the different U^i 's move altogether. The presence of the expectation operator in condition (22) reflects that continuation utility acts as an 'imperfect asset' similar to a riskless bond in a stochastic environment.

The necessary first order conditions for the optimal choice of x^i are

$$V'_x(y^i, U^i, x^i) = (1 + r)\xi \geq 0 \text{ for any } i = 1, 2, \dots, N. \quad (23)$$

According to (23), the planner allocates marginal utilities x^i (not consumption) so that marginal costs are equalized across the different states. Notice that constraint (11) only restricts the average level of marginal utility to be delivered to the agent. The concavity of V would imply that a large level of ξ lowers all x^i for any U increasing future consumption c^i . This is the *backloading effect* induced by the incentive constraint on hidden savings.

The optimal choice of c satisfies

$$-1 + \lambda u'_c(c, e) + \mu u''_{ec}(c, e) + (\xi - \phi)u''_{cc}(c, e) = 0. \quad (24)$$

Since $u''_{cc}(c, e) < 0$, in the additive separable case ($u''_{ec}(c, e) = 0$), condition (24) suggests that if $(\xi - \phi) > 0$ the planner may discourage the agent's savings by 'distorting' today's consumption below the level that would be implied by the Pareto weight λ . Notice however, that only the difference between the distortion needed today (ξ) and the one already inherited from the past (ϕ) matters. For example, when $\xi = \phi > 0$ all the required distortions are fully summarized by λ , hence we have the usual condition $\lambda = \frac{1}{u'(c)}$, i.e. the relative Pareto weight equals the ratio between the planner's and the agent's marginal utilities.

We finally have the first order condition for e :

$$\frac{1}{1+r} \sum_i p'_i(e) V(y^i, U^i, x^i) = -\mu \left[u''_{ee}(c, e) + \beta \sum_i p''_i(e) U^i \right] + \xi \sum_i p'_i(e) x^i + (\phi - \xi) u''_{ce}(c, e). \quad (25)$$

The left hand side represents the discounted gains of the planner induced by an additional marginal unit of effort. The first term on the right hand side represents the increase in the effort incentive

costs. If the agent's problem is concave in e , this term is positive. Under fairly general monotonicity conditions, the second term should be negative: an increase in e relaxes the Euler equation restriction since, facing higher expected future income levels, the agent has less incentives to save. This is a *new effect* of this model and will be important to interpret the simulation results below. The last term depends directly on the complementarity between effort and consumption, and in the linear separable case it equals zero.

4.2 Simulation Based Analysis

The target of this section is threefold. First, we show with a simple application how the recursive formulation can be used to simulate the model and characterize the efficient allocation in more detail. Second, we complement the qualitative result of Section 2 by quantifying the welfare gains in our application for various levels of initial wealth and for different degrees of risk aversion. Finally, we use these welfare calculations to put the verification procedure of the first order approach at work by checking ex-post the incentive compatibility of the optimal allocations for some of our examples.

Model Specification We present all our quantitative results for the infinite horizon case. To reduce computational difficulties we assume only two possible realizations of output ($N = 2$ with $y^h > y^l$).²⁹ We define $p(e) = \Pr(y = y^h \mid e)$, and $1 - p(e) = \Pr(y = y^l \mid e)$. Throughout this section, we also assume the following representation for agents' preferences: $u(c, e) = u(c) - v(e)$. Since we will only consider utility functions satisfying the Inada conditions, we will use the fact that both the Euler equation (11) and (12) are satisfied with equality. Notice that in this case concavity implies a one-to-one relationship between consumption c and marginal utility $x = u'(c)$. As a consequence, we can use consumption itself as a state variable, together with the lifetime utility U and the exogenous state y . We set $\beta(1 + r) = 1$ in order to have stationarity in the benchmark case without asymmetric information. For this range of parameters the efficient allocation does not display explosive consumption paths either.

Functional Forms and Parameters Our two-state model can be best interpreted as a stylized job search - job retention model where high effort induces higher probability of finding a job in the unemployment state and high effort increases the probability of retaining a job in the employment

²⁹In AP, we characterize the contract in the two period case with many income realizations.

state. For expositional simplicity we assume that effort affects these two probabilities in exactly the same manner.³⁰

We chose a standard CRRA utility for consumption: $u(c) = \frac{c^{1-\sigma}}{1-\sigma}$; and quadratic disutility of effort: $v(e) = \eta e^2$. The probability of the high output realization is given by $p(e) = 1 - \exp(-\rho e)$. It might be worthwhile to mention that this specification is within the class of preferences and probability functions for which, in AP, we show that the first order approach is justified, in the two period version of the model.

The table below contains the parameter values we used for our calculations. We used two values of risk aversion for our welfare comparisons, and we chose η in such a way that in the full information economy (no information asymmetry either on effort or on savings) we obtained the same optimal allocation with the two risk-aversion parameters. This implies that in the full information allocation the agent faces the same technical rate of substitution between consumption (output) and effort in both risk aversion cases.

Table 1: Parameter Values

	β	σ	ρ	y^l	y^h	η
Low risk aversion	0.99	0.5	0.1	0.1	100	0.5
High risk aversion	0.99	2	0.1	0.1	100	0.0067

Numerical Methodology The first step of our numerical procedure is to compute the domain restriction M^* . In order to construct M^* , we used a modification of the algorithm proposed by Chang (1998).³¹ We defined a two dimensional grid $M_0 = \{U_k\}_{k=1}^n \times \{c_j\}_{j=1}^m$ where c_j ranged between $\underline{c} = \underline{\alpha}y^l$ and $\bar{c} = \bar{\alpha}y^h$ and U_k ranged between $\underline{U} = \frac{u(\underline{c}) - v(\max(E))}{1-\beta}$ and $\bar{U} = \frac{u(\bar{c}) - v(0)}{1-\beta}$. We then applied the set valued operator $\mathbf{B}$ defined in (13) on M_0 iteratively until we obtained a fixed point. We then approximated the value function over M^* using Chebyshev polynomials. Notice, that according to Proposition 2, given V the *period zero values* for c and U are obtained by solving (7).

The Domain Restriction For the practical implementation of the iterative procedure that determines M^* we used as a benchmark $\underline{\alpha} = 0.1^{-4}$, $\bar{\alpha} = 1$ and r such that $\beta(1+r) = 1$.

³⁰If asset holdings were observable, our model could be seen as an *i.i.d.* shocks version of Zhao's (2001) UI model (who in turn builds on Hopenhayn and Nicolini, 1997).

³¹The details of the procedure are explained in Appendix C1. We should mention that Judd *et al.* (2003), and Cronshaw (1997) provide different numerical techniques to compute sets analogous to M^* . Note, however, that as opposed to our procedure, their methodology works only if M^* is convex.

Figure 1A here

Figure 1A shows our resulting domain restriction set and a set of optimally chosen (U, c) couples for 1000 time periods for the low risk aversion case.³² Notice that the realizations are in the interior of the set and far from the upper limit of consumption. In fact, we checked robustness by increasing $\bar{c}$ and we found that, although the feasible set might expand upwards significantly, the realized equilibrium path does not change in any noticeable way. The set is a relatively nicely shaped and connected set. In particular, and in contrast with the results of some other authors for different models (see Chang, 1998), it does not present ‘holes’.³³

At the top left of the grid, one can notice that high promised consumptions are not compatible with low levels of lifetime utility. This is so since if today’s consumption level is high and at the same time lifetime utility must be kept low, future consumption must decrease; as a consequence agents would have incentives to save, and this cannot be incentive compatible (i.e. it violates the Euler equation). Figure 1B shows that M^* shrinks when the interest rate r increases. This is an intuitive result. Since agents must be discouraged from accumulating assets, when r increases incentive costs increase as well, and therefore the planner is able to implement only a smaller set of allocations.

Figure 1B here

4.2.1 Benchmark Models

In this section, we study the qualitative and quantitative differences between our model, the model of self insurance (bond economy), the full-information case, and the standard moral hazard model (‘pure’ moral hazard, with observable assets).

Self Insurance. Note first, that the self insurance allocation is a benchmark for our model for two reasons. First, it is clearly an incentive compatible allocation with all transfers identically set to 0. Moreover, as it was shown by ACK, in models of hidden information and hidden savings, this is the only incentive compatible allocation. Within the dynamic programming framework, an agent’s problem in the bond economy can be completely described in terms of the properties of the

³²We show the $\sigma = \frac{1}{2}$ case only for expositional convenience. The domain restriction set for the high risk aversion case is available on request.

³³However, we also find that the procedure applied to very coarse grids does artificially eliminate some points in the interior of the set.

value functions U^{Bond} representing the maximal value of lifetime utility that an agent with wealth level z can get. It is recursively defined as

$$U^{Bond}(z_t) = \sup_{b_t \geq y^l/r, e_t \in E, c_t \geq 0} u(c_t) - v(e_t) + \beta \sum_i p_i(e_t) U^{Bond}(z_{t+1}^i); \quad (26)$$

s.t. : $c_t + b_t \leq z_t$; and $z_{t+1}^i = y_{t+1}^i + (1+r)b_t$.

The first constraint is the usual budget restriction, while the second equality represents the law of motion for the agent's total wealth z (the only state). Note, that the 'natural' debt limit $b \geq -\frac{y^l}{r}$ is imposed each period (Aiyagari, 1994). Notice that this problem can be seen as the endogenous labor supply extension of the (permanent-income) model studied by Bewley (1977), and Huggett (1993). When e is not effective - $p_i(e) \equiv \bar{p}_i \forall i$ - we are back to their case. For future reference, recall that one key characteristics of the self insurance model is given by the usual Euler equation:

$$u'(c_t) \geq (1+r)\beta \sum p_i(e_t) u'(c_{t+1}^i),$$

which takes essentially the same form as the incentive condition (5). In order to solve for the optimal allocation in this economy, the method of value function iterations using Chebyshev polynomials was applied again with $z = y + (1+r)b$ as a state variable.³⁴

Pure Moral Hazard. The other key benchmark for our model is the repeated moral hazard model with observable assets. Following the standard recursive contracts literature, we also applied value function iterations to solve the following recursive standard moral hazard model, using lifetime utility U as the only endogenous state variable:³⁵

$$W(U, y) = \max_{c \geq 0, e \in E, U^h, U^l \in D^*} y - c + \frac{1}{1+r} \left[p(e)W(y^h, U^h) + (1-p(e))W(y^l, U^l) \right]$$

s.t. :

$$U = u(c) - v(e) + \beta \left[p(e)U^h + (1-p(e))U^l \right] \quad (27)$$

$$v'(e) = \beta p'(e) (U^h - U^l). \quad (28)$$

Rogerson (1985a) has shown, that in the general additive separable case, a key characteristic of

³⁴For comparison purposes, we imposed the same upper limit on consumption as in the model of moral hazard with hidden savings. This restriction required to impose an upper bound on asset holdings around 10,000.

³⁵In this case, the domain restriction set takes a very simple form: $D^* = \left[\frac{u(e)}{1-\beta}, \frac{u(\bar{c})}{1-\beta} \right]$.

an optimal interior contract is summarized by

$$\frac{1}{1+r} \sum_i p_i(e_t) \frac{1}{u'(c_{t+1}^i)} = \beta \frac{1}{u'(c_t)}. \quad (29)$$

This condition can also be obtained by combining (22) and (24) (together with the envelope) and set $\xi_t = \phi_t = 0$ for all t , i.e. disregarding the incentive constraint on savings.

Full Information. Finally, in the full information allocation, the planner does not face any informational asymmetries regarding effort or savings. Since we do not have aggregate uncertainty here, this allocation involves a deterministic path for c_t and e_t , with

$$\frac{v'(e_t)}{u'(c_t)} = \frac{p'(e_t)(y^h - y^l)}{1+r}.$$

4.2.2 Time Series Properties of the Optimal Contract

First, we focus on the intertemporal properties of consumption, effort, life-time utility and asset holdings in the four models. On Figure 2 and 3, we present the average of 1000 independent simulations, each 500 periods long, for the same four models (full information, pure moral hazard, self insurance, and moral hazard with hidden assets) in the high risk aversion case. For all cases, U_0 was chosen in a way that the planner makes zero profits ($b_0 = 0$ in the bond economy) given the same initial output distribution. These figures can be interpreted in two different ways. First, they represent the expected pattern of consumption, effort, life-time utility and asset balances for a given agent at time 0. Second, by the law of large numbers, this pattern also represents the longitudinal path of the average (or aggregate) levels of the same variables in an economy with many ex ante identical agents.

Figure 2 here

Figure 2 shows the consequences of the introduction of different informational problems on the optimal allocation. We know that in the case without incentive problems, $\beta(1+r) = 1$ implies that the planner implements a constant consumption/effort path. When the planner faces moral hazard with respect to effort alone, the full information allocation cannot be implemented and the result is a decreasing average consumption path. This consumption pattern confirms the martingale

property of the inverse of the marginal utility displayed in equation (29),³⁶ and contributes to the well-known gradual immiseration, or forever decreasing lifetime utility, which is apparent in Figure 2.³⁷ On the other hand, notice that the inverse $(1/x)$ is a strictly convex transformation hence condition (29) is not compatible with the incentive condition (5) because Jensen inequality implies

$$u'(c_t) < \beta(1+r) \sum_i p_i(e_t) u'(c_{t+1}^i),$$

that is, the pure moral hazard allocation is not incentive compatible in our framework. The optimal contract in the standard moral hazard tends to frontload transfers, so that agents would be *willing to save*. The planner is hence forced to adjust the allocation in order to be compatible with the agents' Euler equation (5).

This adjustment changes the time series characteristics of the optimal contract dramatically. In particular, the allocation with hidden assets not only fails to display the tendency towards misery, since u' is a convex function it actually inherits the increasing pattern of consumption and lifetime utility, key properties of the bond economy.

The intertemporal profile of effort is practically the mirror image of the pattern of consumption. Implemented effort responds to two main forces: effort compensation and incentive costs. Since u is a concave function, as c increases it is increasingly costly for the planner to compensate the agent's effort cost with higher consumption, as a consequence e tends to decrease. This wealth effect is reinforced by the fact that for $\sigma \geq 1/2$ also the incentive costs are increasing with U .³⁸

In Figure 3, we compare the average time series pattern of our model with that of the bond economy. Qualitatively, these patterns have some important similarities. However, the constrained efficient allocation allows for a much higher degree of consumption smoothing than self insurance does. In the bond economy the agents tend to (over)accumulate wealth and postpone consumption essentially for precautionary motives. High levels of wealth allow them to both stay away from the borrowing limit and insure themselves against future idiosyncratic risk. Moreover, when they have low levels of assets, they need to use their effort to 'buy' some extra insurance: with higher effort

³⁶In the high risk aversion case (with $\sigma = 2$ and $\beta = \frac{1}{1+r}$) (29) becomes $c_t^2 = \sum_i p_i(e) (c_{t+1}^i)^2$, which implies $c_t > \sum_i p_i(e) c_{t+1}^i$ by Jensen inequality.

³⁷To be precise, we document here that consumption decreases on *average*; Green (1987) and Thomas and Worrall (1990) show that in the ACK framework with monitorable assets this pattern becomes eventually true *almost surely* for any consumption path.

³⁸In Pavoni and Violante (2004) it is formally shown that $\sigma \geq 1/2$ guarantees that incentive costs increase with U in the case with monitorable assets. We conjecture this to be true for the case with hidden assets as well.

they can obtain higher expected income available for wealth accumulation. This explains the much higher initial average effort levels in the bond economy.

In the case of moral hazard with unobservable assets the planner can alleviate both problems by reducing the precautionary saving motive. First, by providing transfers to low-income households the borrowing constraint is relaxed and therefore low wealth agents are allowed to enjoy higher consumption. Second, agents face a less risky income path; hence they have less incentive to save against these income fluctuations. That is why we observe a much flatter consumption path and more stable effort decisions.

Figure 3 here

The forces operating in the hidden asset moral hazard economy place the average consumption growth patterns in-between self insurance and pure moral hazard.³⁹

The asset holdings dynamics on Figures 2 and 3 complete the picture. In the bond economy, agents accumulate assets very heavily in early periods due to precautionary saving motives. Therefore they achieve high levels of lifetime utilities on average much faster than agents in the constrained efficient allocation. At the same time, as agents become wealthier, they exert a smaller and smaller effort, which results in a decreasing level of average output. Therefore, they rely more and more on their interest income for financing their consumption. This explains why in Figure 3 the asset accumulation path becomes flatter. This is a force specific to our model since it is coming through the channel that income is dependent on wealth through effort (leisure) choice.⁴⁰ This mechanism leads to a long run situation where agents in self insurance are on average ‘wealthy and lazy,’ generating an aggregate production level well below that required by (constrained) efficiency.⁴¹

Finally, notice that the asset accumulation pattern in the optimal contract⁴² contrasts sharply with that of pure moral hazard, where the average asset holdings decrease as consumption is frontloaded. The fourth panel of Figure 2 hence suggests a negative cross-country correlation

³⁹In the case of $\sigma < 1$, the same order holds. The average consumption level in the bond economy increases faster than in our model, which in turn displays a pattern of consumption more backloaded than in the standard moral hazard model. The only difference is now that consumption never decreases on average, not even when assets are fully observable.

⁴⁰This effect is for example not present in the standard models of self-insurance with heterogeneous agents (e.g. Huggett, 1993; Aiyagari, 1994; or Krusell and Smith, 1998) since in these economies, agents face exogenous shocks and supply a fixed amount of labour.

⁴¹After 500 periods, the average output in our economy is 0.5 percent below the level of full information, while it is 28.1 percent below in the case of the bond economy.

⁴²To see how we could determine the implied asset holdings in the planner problems see Appendix C2.

between capital flows ($b_{t+1} - b_t$) and informational asymmetries on asset accumulation: economies where the credit market is largely monitored tend to run (trade) deficits in the short run and trade surpluses in the long run. These surpluses offset only partially the interests payments on the accumulated debt, hence the country become more and more indebted. The opposite is true for economies with non-monitored asset markets. It would be interesting to see what happens if these two types of economies are engaged in trading risk-free bonds in a general equilibrium model. This model could provide a new mechanism to explain why capital flows are small from the developed world (monitored asset markets) to the developing countries (non-monitored savings and storage) (Lucas, 1990).

Figure 4A and 4B here

Now, we move on by analyzing the dynamics of consumption dispersion in these different environments. Figure 4A shows the average path of the unconditional standard deviation for consumption in the three models (under full information there is obviously no consumption dispersion). All these models display increasing consumption variation. This is the expected consequence of the ‘history dependence’ present in the different allocations.⁴³ In Figure 4B, we also plot the time pattern of the coefficient of variation for the three models (the standard deviations divided by the mean for each period).

From this figure, it is clear the higher standard deviation in the hidden asset case is not the consequence of the decreasing average consumption in the standard moral hazard. The main underlying reason behind this pattern is that optimal incentive provision requires *higher conditional consumption dispersion* in the efficient allocation than in the observable asset case. In Figure 5, we plot the policies for the different models as a function of the lifetime utility. Looking at the difference between c_h and c_l (as a measure of conditional consumption dispersion), one note that in the relevant range of utilities (i.e., between -10 and -6) this consumption gap is higher for the hidden asset case than for the standard moral hazard. Intuitively, when the agent has secret access to the credit market the relevant (joint) deviation is to shirk and increase savings. By the concavity of u , higher savings deter effort incentives as they reduce the next period covariance between income and utility. This implies that, for similar levels of consumption a larger dispersion is required in

⁴³The initial decreasing pattern in the bond economy is due to the fact that agents’ starting asset position is close to the borrowing limit. As we will see below in Figure 5, this implies a huge dispersion of consumption and high effort for low levels of assets.

equilibrium to satisfy the effort incentive constraint in the efficient allocation compared to the standard moral hazard.⁴⁴

Figure 5 here

This effect is amplified by the implemented effort choice. Figure 5 shows that the planner implements higher effort in the hidden asset case for most utility levels. This might seem to be surprising for two reasons. First, the required higher consumption dispersion implies less efficient insurance provision. Second, CRRA utility functions imply positive precautionary motive for savings, hence consumption dispersion tends to increase the cost of deterring the agent from asset accumulation. On the other hand, we discussed after condition (25) that when assets are unmonitored effort has also a beneficial role for incentives. By making income realizations associated with higher consumption levels more likely to happen a higher effort level relaxes saving incentives. Figure 5 suggests that, under our parametrizations, the latter effect dominates.

4.2.3 Quantifying the Welfare Gains

Proposition 1 states that our allocation improves on welfare compared to the self-insurance allocation. Since the pure moral hazard allocation is not incentive compatible when asset accumulation is not observable, it is also clear that the efficient allocation is Pareto inferior compared to the observable asset case. In this section, we quantify these gains and losses for the stylized application studied above. The emphasis here should not be on the welfare gains per se, but rather on the relative magnitude of the gain compared to self-insurance and the loss compared to the pure moral hazard model. Recall again, that in the ACK case the loss dominates any potential gain in an extreme way. This welfare calculation will also provide the basis for the ex post verification procedure of the first-order condition approach.

Table 2 shows the results of our calculations for welfare gains for the full information, pure moral hazard, and hidden savings cases compared to the bond economy for different interest rates, for different risk aversion parameters and for different initial asset levels. The first panel contains the efficiency gains for the high risk aversion case and for $\beta(1 + r) = 1$.

Table 2 here

⁴⁴This intuition can actually be formally shown in a model where the effort margin can be perfectly controlled such as the case with two effort levels (e.g., Bisin *et al.*, 2004).

We considered an initial distribution of income degenerate at $y_1 = y^l$. In each panel, the first row show gains in terms of the level of initial utility (U_0). These are the utility levels which would guarantee zero total surplus (profits) to the planner. We also calculated compensating differentials in terms of consumption for the three problems and reported them in the second row of each panel. In order to obtain these results, we fixed the policy function of effort to the one of the bond economy.⁴⁵ The first panel shows that welfare gains in our model can be expressed by a 12.9 percentage change in consumption in all states and time periods compared to self insurance. Welfare gains are more than double than the loss compared to the pure moral hazard case ($19.2 - 12.9 = 6.3$ percentage).

We saw, that agents in the bond economy experience a very steep asset accumulation path initially because they are close to their borrowing limit. It is hence important to see how much of the total welfare gains are due to the longer run behavior and how much are due to the fact that the planner can avoid this phase of intensive asset accumulation by relaxing the borrowing limit. For this reason, we also computed the welfare gains for the level of initial assets (life-time utility) such that first period average consumption in the bond economy equals that in the efficient allocation. By this choice of initial assets, we implicitly guarantee that the effective borrowing limits in the two economies are compatible. Since agents in the bond economy can now self insure much better, efficiency gains are reduced, however our economy still significantly improves on the bond economy allocation (2.6 percent in terms of consumption growth), especially compared to the small 0.9 loss due to the additional incentive problems. One important implication of this result is, that although a significant part of the overall welfare gains is coming from the fact that the agent is facing a relaxed borrowing constraint, there are other key remaining sources of efficiency improvements. In the efficient allocation, the planner is still able to provide more insurance across states (see Figure 5) and implement higher effort levels (see Figure 3) and consequently experience higher aggregate output levels.⁴⁶

We also checked how these measures of welfare gains change when we decrease the risk aversion of the agents. Here we have two key observations. For $b_0 = 0$, welfare gains tend to be lower in this

⁴⁵Notice that this normalization reinforces the interpretation of this compensating variation, since this means that we express all welfare gains in terms of consumption changes.

⁴⁶In Abraham and Pavoni (2004a) we decompose welfare gains by their two main sources. In the case of the higher asset levels, insurance and production efficiency contribute in a roughly equal manner to total welfare improvements. However, their relative importance is not uniform across time periods. Since agents are born with no wealth, initially the dominating effect is risk sharing. As time passes, welfare gains coming from insurance decrease and production efficiency tends to dominate.

case because less risk averse agents benefit less from the relaxation of borrowing constraints. After the initial phase of intensive asset accumulation in the bond economy, welfare gains are instead quantitatively more similar in the two risk aversion cases.

More interestingly, the losses due to the presence of hidden savings are increasing in σ . Intuitively, these welfare losses are closely related to the mismatch between the intertemporal patterns of consumption in the two environments. From the Euler equation, a more convex marginal utility generates a more increasing consumption pattern in the efficient allocation. On the other hand, from equation (29), the degree of convexity of the *inverse* of the marginal utility determines how steeply consumption decreases in the standard moral hazard model. It turns out that for CRRA utilities, the degree of convexity of both the marginal utility and the inverse of the marginal utility are increasing in σ . As a consequence, when risk aversion is low ($\sigma = 1/2$) the two patterns are more closely aligned to each other compared to those of $\sigma = 2$, implying that the planner implements more moderate changes compared to the pure moral hazard in this case.

Finally, we document a non-monotonic relationship between the exogenous risk free rate r and the efficiency gains with respect to self insurance. That a reduction in the interest rate increases the welfare gains of the efficient contract compared to the self insurance is intuitive. In the bond economy agents are worse off as they can obtain less self insurance. At the same time the asset accumulation incentive constraint in the planner problem is relaxed by a decrease in the return on savings. In contrast, according to Figure 1B, the state space became smaller when we increased the interest rate, because it is harder to control the agents' incentives to save. Moreover, when r is high self insurance itself performs better compared to the benchmark case. So the increase in welfare gains in this latter case is somehow surprising. However notice that, in contrast to the bond economy, in a constrained efficient allocation the agent has access to a complete set of contingent commodities. An increase in r reduces the (discounted) price of such history contingent commodities, and the agent can make additional profits with respect to self insurance by intertemporally trading assets related to these goods. In particular, when r is high, intertemporal trade gains are obtained by postponing consumption. In other terms, the agent is better off since a higher r reduces the price of insurance. This latter force locally (in the neighborhood of $\frac{1}{1+r} = \beta$) dominated the other ones, and we see increasing welfare gains in this direction as well. This result differs from (and complements) that obtained by Doepke and Townsend (2004). In Doepke and Townsend the welfare gains are induced by the mismatch between the agent's and the planner's intertemporal trade technology. In our comparative static exercise, we maintain the assumption that the planner

and the agent face the same interest rate.

4.2.4 Implementing the Ex-post Verification Procedure

We can now use the numerical solution to implement the verification procedure of the first order condition approach described in Section 3.3. Recall, that the validity of this approach is a prerequisite of our recursive reformulation of the constrained efficient problem.

We first discretize the optimal policy rules (16) and (17) over a grid M . We then define a grid $G = \{b_s\}_{s=1}^q$ on asset holdings and solve (18) as a discrete dynamic programming problem. Notice that by fixing the grid G for b to the singleton $\{0\}$ we could test whether the agent has incentives to deviate when he can choose only his effort level, as consumption is determined by the budget constraint.⁴⁷

Recall that in the separable case we used c as a state. Therefore, using agent's budget constraint the procedure computes the limit point (as $t \rightarrow \infty$) of the sequence of functions

$$V_{t+1}^R(U, c, b) = \max_{b^R \in G, e^R \in E} u(c + (1+r)b - b^R) - v(e^R) + \beta \sum_i p_i(e^R) V_t^R(f(U, c, y^i), h(U, c, y^i), b^R),$$

with $(U, c, b) \in M \times G$, and the first guess of the value function given by

$$V_1^R(U, c, b) = \max_{b^R \in G, e^R \in E} u(c + (1+r)b - b^R) - v(e^R) + \beta \sum_i p_i(e^R) f(U, c, y^i).$$

Note that both the optimal value $U^*(U_0)$ and the laws of motions for U and c are approximated. Moreover, discretizing the state space adds an additional error to their approximation, most importantly, it implies that the agents faces a transfer scheme, which is just an approximation of the 'true' transfer scheme. Therefore we cannot expect $U_0^R(\tau, y) = U^*(U_0)$ to hold exactly. But as we will explain below, these are the very errors which identify our verification test.

For our final tests we used the following grid

$$G = [-0.5, -0.25, 0, 0.25, 0.5, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 14, 19, 25, 35].$$

We found small discrepancies between $U_0^R(\tau, y)$ and $U^*(U_0)$ in both risk aversion cases. In particular, for the low risk aversion case, it is below 0.01 percent for both levels of initial utility

⁴⁷This test, however, could never yield a violation of the first-order condition approach, because one can show that the agent's problem is concave in effort under our parametrization.

where we evaluated the welfare gains. The discrepancy is somewhat larger for the high risk aversion case: 0.021 and 0.008 percent for the low and high initial utility cases respectively. However, these values are still significantly below the magnitude of the welfare gains (8.3 and 1.8 percent respectively with $\sigma = 2$). Moreover, when we compared the optimal policies of this re-maximization problem with the relaxed optimal policies no significant discrepancies were detected either.

The magnitude of the gains from ex post deviations is not very sensitive to the range of the grid on asset holdings. Around $b = 0$ however, we had to carefully chose values on G consistently with the grid of the states (U, c) (in particular with the grid we chose for c). Otherwise, when G is too coarse the agent could be excessively constrained by the discrete available levels of savings. The gains from ex post deviations are also quite sensitive to how we discretize the endogenous states (U, c) . In a previous version of the paper (Ábrahám and Pavoni 2004a), we used a geometric grid where the distance between points decreases with their level. For the results reported above, we used the same number of grid points as in the previous working paper but an equally spaced grid. In the $\sigma = 2$ case, this change resulted in a much lower measure of discrepancy. This sensitivity to the particular form of discretization highlights one potential drawback of the discretized implementation of the ex post verification procedure we adopted here.

In Appendix D, using some concepts from statistical hypothesis testing we study the effectiveness of our verification test in the presence of the above approximation errors in the $T = 2$ version of the model. Let's define U_0 as the lifetime utility attainable in self insurance with first period asset holdings b and first period income degenerate at y . Then⁴⁸ $D(U_0) = (U^R(U_0) - U^{Foc}(U_0))/U^{Foc}(U_0)$ represents the percentage utility gain from ex-post deviations, where $U^{Foc}(U_0)$ is the utility obtainable by the agent under the relaxed optimal contract with the same initial resources, that is $V_{foc}^*(U^{Foc}(U_0)) = -b(1 + r)$, and $U^R(U_0) = V_\infty^R(U^{Foc}(U_0), c_1^{Foc}(U_0), 0)$ is the solution of (18) with $c_1^{Foc}(U_0)$ being the optimal first period consumption defined by (7). Recall that our aim is to detect whether $D(U_0) = 0$ in the presence of numerical errors. In order to obtain a deviation measure $R(U_0)$ which is comparable across initial utility entitlements (U_0) and risk preferences, we normalize the percentage gains from ex post deviations by the measured percentage welfare gains relative to self insurance (see Table 2):

$$R(U_0) = \frac{D(U_0)}{WG^{foc}(U_0)} = \frac{U^R(U_0) - U^{Foc}(U_0)}{U^{Foc}(U_0) - U_0} \frac{U_0}{U^{Foc}(U_0)}. \quad (30)$$

⁴⁸We simplify the notation by not indicating the initial income y .

Using the analogy with statistical tests, we could maintain the null hypothesis $R(U_0) \leq 0$ and see whether we can fail to reject it with some ‘significance’ without losing too much ‘power’. Kocherlakota (2004) shows that for the case where agent’s preferences are linear in effort, and effort affects linearly job finding probabilities, the first order approach fails. Since this double linear model is concave in both e and b *alone*, there is a sense in which it is not ‘far’ from the class of models for which we know the first order approach should work. In the Appendix, we construct a class of models where the use of the first order approach cannot be analytically justified by using a parametrized family of perturbations of the double linear model proposed by Kocherlakota.⁴⁹ We then design different grid sizes for both these ‘linear’ models and for globally concave models and for each of those we compute the average value of the realization of our measure $\hat{R}(U_0)$ across different U_0 ’s; and we denote this number by $\hat{R}$. We use the variation of this statistic across different models to evaluate the ex post deviation measures of our infinite horizon models.

In the light of the results presented there, it seems that the first order approach could be justified for both risk aversion cases. For the low initial utility case the computed $\hat{R}(U_0)$ is equal to 0.0007 and 0.0025 for the low and high risk aversion cases respectively. These numbers are much below the values we obtained with non concave models with comparable grid size (see Table D1) and also comparable to the magnitude of the ex post deviations we detected in a globally concave model in the presence of approximation errors with similar grid size for $T = 2$ (see Table D2).⁵⁰

5 Conclusions

Relaxing the assumption of perfect observability and contractability on agents’ asset holdings in the dynamic moral hazard model is a potentially very valuable exercise from both the theoretical and applied point of view. However, the introduction of hidden assets introduces serious methodological complications. In this paper we propose a way of solving the methodological problems, and we study the main characteristics of the optimal contract, both analytically and by solving numerically a simple application.

We show that by using a generalized first order condition approach the model can be solved within the recursive contracts framework where, together with the promised lifetime utility, we

⁴⁹We thank Narayana Kocherlakota for suggesting us to use his model for this exercise.

⁵⁰We calculated $\hat{R}(U_0)$ for several levels of initial utilities obtained in the bond economy. We did not find much variation across initial utility levels. The maximum value for $\hat{R}(U_0)$ was 0.0012 and 0.007 for the low and high risk aversion cases respectively.

used the agent's marginal utility of consumption as an additional endogenous state variable. The recursive formulation permits a parsimonious numerical ex-post verification procedure of the first order approach.

Allowing for hidden asset accumulation changes dramatically the long run properties of the efficient allocation compared to the standard moral hazard model (with observable assets). The immiseration pattern typical of the standard moral hazard model is replaced by properties qualitatively similar to those of self insurance, such as the (on average) increasing pattern of consumption and lifetime utility. We also find that the presence of hidden assets induces a new return for effort. Since high effort levels increase the likelihood of high consumption realizations, it partially relaxes the saving incentive constraint. As a consequence, despite that effort incentives require - *ceteris paribus* - higher consumption dispersion, in our application we find that for all relevant levels of utility, the efficient allocation implements a higher effort level than the standard moral hazard model. As a result both conditional and unconditional consumption dispersion is larger in the efficient allocation than in the standard moral hazard model.

In the paper we also demonstrate that the efficient allocation typically improves on self insurance, even in the case where the agent can secretly borrow and lend. We argue that the key assumption determining previous (ACK's) no welfare gains result is that there the agent has complete control over the outcomes the planner can observe. We show that whenever this assumption is relaxed there exists a social insurance contract improving aggregate welfare. Moreover, we find that welfare gains can be substantial.

The welfare gain result is relevant since it leaves the possibility of government interventions open in examples such as unemployment insurance or other welfare programs, where the presence of informational asymmetries between the government and the agents prevent the implementation of first best policies. The general model and methodology studied in this paper can be applied to study the qualitative and quantitative characteristics of optimal policies for those specific examples. For example, with an appropriate calibration exercise the model can be used to study welfare implications of optimal unemployment insurance schemes in the presence of hidden asset holdings. Our results can be compared to those previously obtained in the literature, where the complication of the model forced the authors to consider suboptimal transfer schemes (e.g. Hansen and İmrohoroglu, 1992; and Abdulkadiroğlu et al. 2002).

A natural extension of this work would be to study the closed economy (general equilibrium) version of our model where the interest rate is determined endogenously. That model would enable

us to study the steady state distribution of consumption. Another interesting application along these lines is the study of the asset/capital flows in a two country general equilibrium environment. Our results in Section 4.2.2 suggest that these flows are intimately related to the extent how much governments can observe private asset accumulation decisions.

The framework we develop in this paper can be fruitfully used to study the optimal (private) contract in several other moral hazard problems where hidden savings may be relevant (e.g., long-term employment and compensation contracts, corporate loans and managerial contracts), in a systematic way. More generally, the ex post verification procedure we propose in this paper could be easily applied to a richer set of models. For example, in the recursive formulation of Fernandes and Phelan (2000) and Doepke and Townsend (2004), the set of incentive compatibility constraints easily becomes very large. The computational burden could be reduced significantly by imposing only a carefully chosen (and economically guided) subset of the constraints, and verify ex post whether the obtained efficient allocation is in fact incentive compatible. Another application could be to Ramsey optimal taxation models, when it is difficult to guarantee global concavity of the household's program.⁵¹ In this case, our ex post verification procedure would be equivalent to solving the relevant competitive equilibrium taking the tax rate processes as given.

In terms of the implementation of our ex post verification procedure, we saw that the results may be sensitive to the particular form of discretization of the state space (and consequently the transfer process). Therefore one key improvement of this procedure would be to solve the re-maximization problem by approximating both the transfer process and the value function by continuous functions (e.g., polynomials or cubic splines). However, the better precision would clearly imply much higher computational costs as well.

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⁵¹We have situations in mind where there is a discrete labor market participation decision, or where the planner imposes income tax schemes with some degree of regressivity.

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Appendix A: Proofs

Proof of Proposition 1 Notice that the bond economy allocation can be seen as a planner transfer scheme where period t transfers are h^{t-1} measurable. Consider now the last two periods of the program. To replicate self insurance, the continuation plan

$$\tau \setminus h^{T-1} = \{\tau_{T-1}(h^{T-1}), \tau_T(h^{T-1}, y_T^i)\}_{i=1}^N$$

must be such that $\tau_T(h^{T-1}, y_T^i) = \tau_T^* \forall i$. We now show that a transfer scheme τ_T that does not change with y_T^i cannot be optimal. This will prove our statement since the planner cannot loose with respect to self insurance in previous periods. To simplify notation denote $\tau_{T-1}(h^{T-1}) = \tau_{T-1}^*$. Without loss of generality assume that u is unbounded below and that both 1 and N belong to I , and consider the following modification of the scheme: leave both τ_{T-1}^* and $\hat{\tau}_T^i(\varepsilon) = \tau_T^*$ for $i = 2, \dots, N-1$ unchanged, and set $\hat{\tau}_T^1(\varepsilon) = \tau_T^* + \varepsilon$, and $\hat{\tau}_T^N(\varepsilon) = \tau_T^* - \mu^\varepsilon \varepsilon$, with $\varepsilon > 0$. For any ε , choose μ^ε so that the agent is indifferent,⁵² that is

$$\begin{aligned} & \max_{e \in E, b \geq -B(\varepsilon)} u(y_{T-1} + \tau_{T-1}^* - b, e) + \beta \sum_i p_i(e) u(y_T^i + \hat{\tau}_T^i(\varepsilon) + (1+r)b, 0) \\ &= \max_{e \in E, b \geq -B(\varepsilon)} u(y_{T-1} + \tau_{T-1}^* - b, e) + \beta \sum_{i=2}^{N-1} p_i(e) u(y_T^i + \tau_T^* + (1+r)b, 0) \\ & \quad + \beta [p_1(e) u(y_T^1 + \tau_T^* + \varepsilon + (1+r)b, 0) + p_N(e) u(y_T^N + \tau_T^* - \mu^\varepsilon \varepsilon + (1+r)b, 0)] \\ &= u(y_{T-1} + \tau_{T-1}^*, e_{T-1}^*) + \beta \sum_i p_i(e_{T-1}^*) u(y_T^i + \tau_T^*, 0) \quad (\text{recall that } b_{T-1}^* = 0), \end{aligned}$$

where for each ε , $B(\varepsilon) = \frac{1}{1+r} \min \{y_T^i + \hat{\tau}_T^i(\varepsilon)\}$. Such a μ^ε exists by the Maximum Theorem. Notice indeed that both u and $\mathbf{p}$ are continuous and E is compact. Moreover, without loss of generality, we can require $b \in B(\varepsilon, \mu)$, where for any μ and ε

$$B(\varepsilon, \mu) \equiv [-\frac{1}{1+r} \min \{y_T^1 + \tau_T^* + \varepsilon, y_T^2 + \tau_T^*, y_T^N + \tau_T^* - \mu^\varepsilon \varepsilon\}, y_{T-1} + \tau_{T-1}^*]$$

is a non-empty, compact and continuous correspondence. Finally, note that for any $\varepsilon > 0$ if $\mu = 0$ the utility of the agent must be strictly larger than the one in the original contract; since u is unbounded and the fact that $p_1(e), p_N(e) > 0$ for any e , by reducing μ we can make the utility of the agent arbitrarily low. Hence, by continuity there must be a μ^ε that satisfies our requirement.

If we now denote by $\hat{e}_{T-2}^\varepsilon$ the effort taken by the agent under the new scheme, the difference in the

⁵²When u is unbounded above, we shall modify the original self insurance scheme as follows: $\hat{\tau}_{T-1} = \tau_{T-1}^*$, $\hat{\tau}_T^i(\varepsilon) = \tau_T^*$ for $i = 2, \dots, N-1$, $\hat{\tau}_T^1(\varepsilon) = \tau_T^* + \mu^\varepsilon \varepsilon$, and $\hat{\tau}_T^N(\varepsilon) = \tau_T^* - \varepsilon$.

returns for the planner between the two schemes is

$$\delta(\varepsilon) = p_N(\hat{e}_{T-1}^\varepsilon)\mu^\varepsilon\varepsilon - p_1(\hat{e}_{T-1}^\varepsilon)\varepsilon.$$

We want to show that for ε small enough $\delta(\varepsilon) > 0$. To this extent we can take the limit as $\varepsilon \rightarrow 0$ of

$$\frac{\delta(\varepsilon)}{\varepsilon} = p_N(\hat{e}_{T-1}^\varepsilon)\mu^\varepsilon - p_1(\hat{e}_{T-1}^\varepsilon).$$

and show that $\delta'(0) = p_N(\hat{e}_{T-1}^0)\mu^0 - p_1(\hat{e}_{T-1}^0) > 0$, where $\hat{e}_{T-1}^0$, $\hat{b}_{T-1}^0$ and μ^0 are the limit effort, asset and μ^ε choices when ε is driven to zero. The Theorem of the Maximum implies that the optimal correspondence of choices $\hat{e}_{T-1}^\varepsilon$ and $\hat{b}_{T-1}^\varepsilon$ is upper hemicontinuous if considered as a function of ε . The consequence is that $\hat{e}_{T-1}^0$ and $\hat{b}_{T-1}^0$ are also optimal under the original scheme, hence for any $\varepsilon > 0$ we have

$$\begin{aligned} & u(y_{T-1} + \tau_{T-1}^* - \hat{b}_{T-1}^0, \hat{e}_{T-1}^0) + \beta \sum_i p_i(\hat{e}_{T-1}^0) u\left(y_T^i + \hat{\tau}_T^i(\varepsilon) + (1+r)\hat{b}_{T-1}^0, 0\right) \\ \leq & u(y_{T-1} + \tau_{T-1}^* - \hat{b}_{T-1}^\varepsilon, \hat{e}_{T-1}^\varepsilon) + \beta \sum_i p_i(\hat{e}_{T-1}^\varepsilon) u\left(y_T^i + \hat{\tau}_T^i(\varepsilon) + (1+r)\hat{b}_{T-1}^\varepsilon, 0\right) \\ = & u(y_{T-1} + \tau_{T-1}^* - \hat{b}_{T-1}^0, \hat{e}_{T-1}^0) + \beta \sum_i p_i(\hat{e}_{T-1}^0) u\left(y_T^i + \tau_T^* + (1+r)\hat{b}_{T-1}^0, 0\right), \end{aligned}$$

or

$$\sum_i p_i(\hat{e}_{T-1}^0) \left[u\left(y_T^i + \hat{\tau}_T^i(\varepsilon) + (1+r)\hat{b}_{T-1}^0, 0\right) - u\left(y_T^i + \tau_T^* + (1+r)\hat{b}_{T-1}^0, 0\right) \right] \leq 0. \quad (31)$$

Since u is concave it is superdifferentiable, hence (31) implies that

$$\begin{aligned} 0 & > p_1(\hat{e}_{T-1}^0)\varepsilon\partial_1 u\left(y_T^1 + \tau_T^* + (1+r)\hat{b}_{T-1}^0, 0\right) \\ & \quad - p_N(\hat{e}_{T-1}^0)\mu^\varepsilon\varepsilon\partial_1 u\left(y_T^N + \tau_T^* + (1+r)\hat{b}_{T-1}^0 - \mu^\varepsilon\varepsilon, 0\right), \end{aligned}$$

where $\partial_1 u(c, 0)$ is any superdifferentiable of u with respect to the first argument at c . Dividing by ε and taking the limit $\varepsilon \rightarrow 0$ we get

$$0 \geq p_1(\hat{e}_{T-1}^0)\partial_1 u\left(y_T^1 + \tau_T^* + (1+r)\hat{b}_{T-1}^0, 0\right) - p_N(\hat{e}_{T-1}^0)\mu^0\partial_1 u\left(y_T^N + \tau_T^* + (1+r)\hat{b}_{T-1}^0, 0\right). \quad (32)$$

It is now easy to realize that by strict concavity

$$\partial_1 u\left(y_T^1 + \tau_T^* + (1+r)\hat{b}_{T-1}^0, 0\right) > \partial_1 u\left(y_T^N + \tau_T^* + (1+r)\hat{b}_{T-1}^0, 0\right)$$

hence (32) implies $\delta'(0) > 0$. **Q.E.D.**

The last steps of the proof borrow from and generalize Grossman and Hart (1983), Proposition 5.

Proof of Proposition 2 In what follows, without loss of generality we will consider contracts where $\mathbf{b}_t(h^t) \equiv 0$. Notice that to each contract $\mathcal{W}$, we can associate the states as follows:

$$u'_c(\mathbf{c}_t(h^t), \mathbf{e}_t(h^t)) = \mathbf{x}_t(h^t) \quad (33)$$

$$\mathbf{U}_t(\mathcal{W}; h^t) = \mathbf{U}_t(h^t). \quad (34)$$

From the definition of lifetime utility $\mathbf{U}_t(\mathcal{W}; h^t)$, (34) satisfy the following version of (9)

$$\mathbf{U}_t(h^t) = u(\mathbf{c}_t(h^t), \mathbf{e}_t(h^t)) + \beta \sum_i p_i(\mathbf{e}_t(h^t)) \mathbf{U}_{t+1}(h^t, y^i); \quad (35)$$

and - by construction - (33) guarantees the sequential version of (12)

$$u'_c(\mathbf{c}_t(h^t), \mathbf{e}_t(h^t)) \leq \mathbf{x}_t(h^t). \quad (36)$$

Moreover, if the resulting plan of states is generated by an incentive feasible contract $\mathcal{W} \in \Omega_{FOC}$ then by definition $\mathbf{c}_t(h^t) \geq 0$ and $\mathbf{e}_t(h^t) \in E$ and it must satisfy the following sequential version of (10), (11) and the domain restriction

$$-u_e(\mathbf{c}_t(h^t), \mathbf{e}_t(h^t)) = \beta \sum_i p'_i(\mathbf{e}_t(h^t)) \mathbf{U}_{t+1}(h^t, y^i) \quad (37)$$

$$u'_c(\mathbf{c}_t(h^t), \mathbf{e}_t(h^t)) \geq \beta(1+r) \sum_i p_i(\mathbf{e}_t(h^t)) u'_c(\mathbf{c}_{t+1}(h^t, y^i), \mathbf{e}_{t+1}(h^t, y^i)) \quad (38)$$

$$(\mathbf{x}_t(h^t), \mathbf{U}_t(h^t)) \in M^* \quad (39)$$

The domain restriction is satisfied since whenever $\mathcal{W} \in \Omega_{FOC}$, the contract in period zero generates incentive compatible continuations $\mathcal{W} \setminus y^i$.

Denote by $\mathcal{S} = \{\mathbf{x}_t(h^t), \mathbf{U}_t(h^t)\}_{t=1}^\infty$ the contingent plan of states and by $\mathcal{M}^*$ the set of contingent plan of states and contracts that satisfy all such constraints, i.e.

$$\mathcal{M}^* = \{(\mathcal{S}, \mathcal{W}) : \text{for all } h^t \mathbf{c}_t(h^t) \geq 0, \mathbf{e}_t(h^t) \in E, (35)-(36)-(37)-(38)-(39)\}.$$

We have just shown that for each $\mathcal{W} \in \Omega_{FOC}$ we can find a pair $(\mathcal{S}, \mathcal{W}) \in \mathcal{M}^*$. We now want to show the converse. Notice first that from the definition of M^* , for each continuation $\mathbf{U}_t(h^t)$ utility in $(\mathcal{S}, \mathcal{W}) \in \mathcal{M}^*$ there is an incentive contract $\mathcal{W}$ such that $\mathbf{U}_t(\mathcal{W}; h^t) = \mathbf{U}_t(h^t)$. As a consequence, (35) together with the domain restriction guarantee that we generate well defined values for the agent. Moreover, (37) together with the domain restriction implies that effort incentive compatibility (4) is satisfied at each node. In order to show that (11) implies that the saving incentive compatibility is satisfied we need to show that (36) and

(38) together guarantee (5). This is the case, because for any two consecutive periods at node h^t we have

$$\begin{aligned} u'_c(\mathbf{c}_t(h^t), \mathbf{e}_t(h^t)) &\geq \beta(1+r) \sum_i p_i(\mathbf{e}_t(h^t)) \mathbf{x}_{t+1}(h^t, y^i) \\ &\geq \beta(1+r) \sum_i p_i(\mathbf{e}_t(h^t)) u'_c(\mathbf{c}_{t+1}(h^t, y^i), \mathbf{e}_{t+1}(h^t, y^i)) \end{aligned}$$

which is just (5).

By the above argument problem (6) can be equivalently written as

$$\begin{aligned} V_{foc}^*(U_0) &= \sup_{(\mathcal{W}, \mathcal{S}) \in \mathcal{M}^*} \sum_i p_i^0 \Pi_1(\mathcal{W}; y^i) \\ \text{s.t. } &\sum_i p_i^0 \mathbf{U}_1(\mathcal{W}; y^i) \geq U_0. \end{aligned}$$

It is now a straightforward application of the Bellman principle to show that the true value of the problem V_{foc}^* can be decomposed as follows:

$$\begin{aligned} V_{foc}^*(U_0) &= \sup_{(x^i, U^i) \in M^*} \sum_i p_i^0 W_{foc}^*(y^i, x^i, U^i) \\ \text{s.t. } &: \sum_i p_i^0 U^i \geq U_0, \end{aligned} \tag{40}$$

where, for $(x^i, U^i) \in M^*$ and $y^i \in Y$, we have

$$\begin{aligned} W_{foc}^*(y^i, x^i, U^i) &= \sup_{(\mathcal{W}, \mathcal{S}) \setminus y^i \in \mathcal{M}^*} \Pi_1(\mathcal{W}; y^i) \\ \text{s.t. } \mathbf{U}_1(y^i) &= U^i; \mathbf{x}_1(y^i) = x^i. \end{aligned}$$

The proof of this last statement requires to verify for the last problem the properties for the sup operator which defines the original function $V_{foc}^*(U_0)$. To show that it is an upper bound notice that since any pair $(\bar{\mathcal{W}}, \bar{\mathcal{S}}) \in \mathcal{M}^*$ such that $\sum_i p_i^0 \mathbf{U}_1(\bar{\mathcal{W}}; y^i) \geq U_0$ induces continuations $(\bar{x}^i, \bar{U}^i) \in M^*$, it follows that $\bar{\mathbf{U}}_1(y^i) = \bar{U}^i$; $\bar{\mathbf{x}}_1(y^i) = \bar{x}^i$ and that $W_{foc}^*(y^i, \bar{x}^i, \bar{U}^i) \geq \Pi_1(\bar{\mathcal{W}}; y^i)$. We must hence have that $\sup_{(x^i, U^i) \in M^*} \sum_i p_i^0 W_{foc}^*(y^i, x^i, U^i) \geq \sum_i p_i^0 W_{foc}^*(y^i, \bar{x}^i, \bar{U}^i) \geq \sum_i p_i^0 \Pi_1(\bar{\mathcal{W}}; y^i)$ for all such pairs. We now want to show that it is the least upper bound. Take any $\varepsilon/2 > 0$. By the definition of sup in (40) there exists a set of pairs $(\bar{x}^i, \bar{U}^i) \in M^*$ $i = 1, 2, \dots, N$ such that $\sum_i p_i^0 \bar{U}^i \geq U_0$ and $\sup_{(x^i, U^i) \in M^*} \sum_i p_i^0 W_{foc}^*(y^i, x^i, U^i) < \sum_i p_i^0 W_{foc}^*(y^i, \bar{x}^i, \bar{U}^i) + \varepsilon/2$. Moreover, for any i and $(\bar{x}^i, \bar{U}^i) \in M^*$ there exists a pair $(\bar{\mathcal{W}}, \bar{\mathcal{S}}) \setminus y^i \in \mathcal{M}^*$ such that $\bar{\mathbf{U}}_1(y^i) = \bar{U}^i$; $\bar{\mathbf{x}}_1(y^i) = \bar{x}^i$ and $W_{foc}^*(y^i, \bar{x}^i, \bar{U}^i) < \Pi_1(\bar{\mathcal{W}}; y^i) + \varepsilon/2$. All this implies that for any $\varepsilon > 0$ there exists $(\bar{\mathcal{W}}, \bar{\mathcal{S}}) \in \mathcal{M}^*$ such

that $\sum_i p_i^0 \mathbf{U}_1(\bar{\mathcal{W}}; y^i) \geq U_0$ and $\sup_{(x^i, U^i) \in M^*} \sum_i p_i^0 W_{foc}^*(y^i, x^i, U^i) < \sum_i p_i^0 \mathbf{\Pi}_1(\bar{\mathcal{W}}; y^i) + \varepsilon$.

Following a line of proof similar to that we used above and using again standard arguments (e.g., using a direct application of Theorem 4.2 of SLP) it is easy to show that the ‘interim’ value function W_{foc}^* solves the functional equation (8)-(12). The key is to realize that the restriction $\mathcal{M}^*$ can be equivalently written recursively by using the following state dependent correspondence

$$\Gamma(x, U) = \left\{ \left(c, e, \{U^i, x^i\}_{i=1}^N \right) : c \geq 0, e \in E, (U^i, x^i) \in M^* \text{ for all } i \text{ (9)-(10)-(11)-(12)} \right\}$$

which is straightforward to verify directly from the definition of $\mathcal{M}^*$.

The converse is standard. When V is bounded we can use Theorem 4.3 of SLP to show that $V = W_{foc}^*$. In this case, since M^* is compact, we can apply Theorem 4.6 of SLP and prove that V is continuous. Notice that we can use the theorem of the maximum (e.g. see Theorem 3.6 in SLP) despite that the restriction on $c \geq 0$ is unbounded above. This is so since the objective function is *coercive* in c and this allow us to restrict the choice of c to compact sets only.⁵³ The maximum theorem hence guarantees that the policy correspondence is non empty for any $(y, x, U) \in Y \times M^*$, and existence of an optimal plan can be shown by repeatedly applying (any selection of) the policy. **Q.E.D.**

Proof of Proposition 3 Consider the case where p^0 is degenerate at y , and let U_0 the utility to be delivered to the agent in period zero. We denote by $\mathcal{W}_{foc}^*(U_0)$ the relaxed optimal contract (which exists by Proposition 2), and by $V^*(U_0, y)$ the value of the true optimal (fully incentive compatible) contract associated to problem (3). In terms of the proposition, $\mathbf{U}^*(U_0) = \mathbf{U}_1^T(\mathcal{W}_{foc}^*(U_0), y) \geq U_0$. Clearly if $U_0^R(\tau, y) > U^*(U_0)$ there exists a feasible deviation hence $\mathcal{W}_{foc}^*(U_0)$ cannot be optimal since it is not incentive feasible, i.e. $\mathcal{W}_{foc}^*(U_0) \notin \Omega$.

To show the converse, notice first that - since $\Omega \subset \Omega_{FOC} - \mathbf{\Pi}_1^T(\mathcal{W}_{foc}^*(U_0), y) \geq V^*(U_0, y)$. Hence, whenever $\mathcal{W}_{foc}^*(U_0) \in \Omega$ then $\mathcal{W}_{foc}^*(U_0)$ is optimal. Moreover, since obeying to the proposed contract is feasible for the agent we always have $U_0^R(\tau, y) \geq U^*(U_0)$. We hence have to show that $U_0^R(\tau, y) = U^*(U_0)$ implies that $\mathcal{W}_{foc}^*(U_0) \in \Omega$, i.e. it is sequentially incentive compatible.

If at node h^t , $\mathcal{W}_{foc}^*(U_0)$ is not incentive compatible, there must be a feasible deviation strategy $\bar{\sigma}$ such

⁵³Since V is bounded we have coercivity: $\lim_{c \rightarrow \infty} f(c) = y - c + \frac{1}{1+r} \sum p_i(e) V(y^i, U^i, x^i) = -\infty$. Coercivity and continuity guarantee that the objective function f has compact upper sections:

$$U(\alpha) = \{c \geq 0 : f(c) \geq \alpha\},$$

and we can without loss of generality focus on the compact sets $U(\alpha)$ for c .

that $\mathbf{U}_t^T(\boldsymbol{\tau}_{foc}^*, \bar{\boldsymbol{\sigma}}; h^t) > \mathbf{U}_t^T(\boldsymbol{\tau}_{foc}^*, \boldsymbol{\sigma}_{foc}^*; h^t)$. Suppose we construct a new plan $\bar{\boldsymbol{\sigma}}_{foc}^*$ for the agent as follows. Assume the agent behaves as suggested by the contract both before period t and after period t in all nodes but h^t ; while at node h^t he follows plan $\bar{\boldsymbol{\sigma}}$. This plan is clearly available to him in the remaximization problem. It must hence be that $U_0^R(\boldsymbol{\tau}, y) \geq \mathbf{U}_1^T(\boldsymbol{\tau}_{foc}^*, \bar{\boldsymbol{\sigma}}_{foc}^*; y) > U^*(U_0)$ where the last inequality is due to the assumption that $\beta > 0$ and that the planner's prescription of the contract $\mathcal{W}_{foc}^*(U_0)$ generates full support, hence each node is reached with positive probability. **Q.E.D.**

Appendix B: A Counter-Example

Consider the two period version of the model presented in Section 2 with two income realizations: $y_2^l = 0 < 1 = y_2^h$, a separable utility function of the form $u(c, e) = \sqrt{c} - e$, $c \geq 0$, $E = [\underline{e}, \bar{e}]$, with $\underline{e} \geq 0$, $\bar{e} \in (\frac{1}{2}, 1)$; and the following linear probability: $p(e) = \Pr\{y = y^h \mid e\} = e$. For simplicity, we set $\beta = 1$ and $r = 0$. We moreover assume that the initial endowment is equal to zero, and agents are born with zero wealth (i.e., $y_1 = b_0 = 0$).

If there is no planner who can provide additional insurance, the agent in the bond economy solves the following problem:

$$U_2^{Bond}(0) = \max_{e_1 \in [\underline{e}, \bar{e}], b_1 \geq 0} u(-b_1) - e_1 + e_1 u(1 + b_1) + (1 - e_1) u(b_1),$$

where b_1 is the first-period asset choice. Obviously, the agent would like to borrow in the first period; however he is forced to have $c_1 = 0$ and will bring no assets into the second period ($b_1 = 0$). Notice that the agent is indifferent between all effort levels since $u(1 + b_1) - u(b_1) = \sqrt{1} - \sqrt{0} = 1$, which implies $U_2^{Bond}(0) = 0$.

On the other hand, if the planner does not face any informational problems he will provide full insurance to the agent. It is easy to see that the unconstrained optimal allocation would be $c_1 = c_2^l = c_2^h = \frac{1}{4}$ and $e_1 = \frac{1}{2}$, guaranteeing the agent a lifetime utility level of $\frac{1}{2}$.

It is now crucial to distinguish between two cases of moral hazard with hidden asset accumulation. (i) Assume first that $\underline{e} > 0$. In accordance with Proposition 1, in this case the planner is able to improve welfare. Although the optimal contract obviously depends on the particular value of $\underline{e}$ notice, that the planner can always make non-negative profits and provide full-insurance with $c_1 = c_2^l = c_2^h = \frac{\underline{e}}{2}$ and $e_1 = \underline{e}$. This allocation is clearly incentive compatible and improves the agent's welfare with respect to self insurance since, by concavity, $2\sqrt{\frac{\underline{e}}{2}} - \underline{e} > 0$. (ii) Assume now that $\underline{e} = 0$. In this situation, one can easily check that the only incentive compatible allocation of consumption is $c_1 = c_2^l = 0$ and $c_2^h = 1$, which is identical to that of the bond economy. This is an example of the ACK self insurance result for our model. Notice that any transfer scheme which provides some additional insurance implies a deficit for the planner since the agent

will exert $e = 0$, and $p(0) = 0$ implies that the taxed (high) outcome will never appear.

Appendix C: Quantitative Analysis

C1. Domain Restriction. Let us rewrite the definitions of the feasible set for our separable case with c as a state variable. The domain restriction set M^* is the set of (U, c) couples such that there exists $(U^l, c^l) \in M^*$, $(U^h, c^h) \in M^*$ and $e \in E$ such that (41)-(43) are all satisfied:

$$U = u(c) - v(e) + \beta (U^l + p(e)(U^h - U^l)) \quad (41)$$

$$v'(e) = \beta p'(e)(U^h - U^l) \quad (42)$$

$$u'(c) = \beta(1+r)(u'(c^l) + p(e)(u'(c^h) - u'(c^l))). \quad (43)$$

We defined a two dimensional grid $M_0 = \{U_k\}_{k=1}^n \times \{c_j\}_{j=1}^m$ where c_j ranged between $\underline{c}$ and $\bar{c}$ and U_k ranged between $(u(\underline{c}) - v(\bar{e})) / (1 - \beta)$ and $\bar{U}$, where $\bar{e} \equiv \max(E)$ and $\bar{U} = (u(\bar{c}) - v(0)) / (1 - \beta)$. Then, we applied the set valued operator $\mathbf{B}$ defined bellow on M_0 iteratively until we obtained a fixed point:

$$M_t = \mathbf{B}(M_{t-1}) = \{(U, c) \in M_{t-1}; \exists (U^i, c^i) \in M_{t-1}, e \in E; (41)-(43) \text{ hold}\}. \quad (44)$$

Conditions (41)-(43) impose 3 constraints on our 5 endogenous variables $[U^l, c^l, U^h, c^h, e]$. Since M_{t-1} is a discrete set, for any $(U, c) \in M_{t-1}$ it is not always possible to have $(U^l, c^l) \in M_{t-1}$ and $(U^h, c^h) \in M_{t-1}$ satisfying (41)-(43) exactly for any $e \in E$. Note, that this has nothing to do with the actual feasibility of these bundles. For this reason, we had to modify the algorithm defined by (44) and use the following operator instead:

$$\tilde{\mathbf{B}}(M_{t-1}) = \left\{ \begin{array}{l} (U, c) \in M_{t-1}; \exists (U^i, c^i) \ i = l, h \text{ and } e \in E \\ \text{such that (41)-(43) are satisfied and} \\ \forall i \exists (U_k^i, c_j^i) \in M_{t-1} \text{ s.t. } \|(U^i, c^i) - (U_k^i, c_j^i)\| < \varepsilon_{j,k} \end{array} \right\} \quad (45)$$

$\tilde{\mathbf{B}}(M_{t-1})$ excludes (U, c) bundles for which there is no such solution of (41)-(43) that has a close enough point in the discrete set M_{t-1} . Generally, we chose $\varepsilon_{j,k}$ being one half of the distance between grid points around (U_k^i, c_j^i) .

We had to modify $\tilde{\mathbf{B}}(\cdot)$ for some practical reasons around the boundaries. First, note that $\underline{c} = 0$ would have raised numerical problems, since in this case the marginal utility of consumption (included in the Euler equation) is equal to minus infinity. For this reason we chose the lowest consumption in the grid being $\underline{c} = 0.0001y^l$. However, we chose $\varepsilon_{j,1}$ in a way that we did not exclude points where the only feasible continuation included $c^l \in (0, \underline{c}]$. The fixed point of this modified operator is given by $\tilde{M}^* = \tilde{\mathbf{B}}(\tilde{M}^*)$. $\tilde{M}^*$ is the discrete approximation of the feasibility set we used throughout the paper. The feasibility sets presented in the main text are the results of this procedure.

C2: Recovering Agents' Asset Holdings. In this section we describe how we computed the agents asset level in the different planner problems. There is an obvious sense in which this exercise also decentralizes the optimal allocation with exclusive contracts.

Notice that to any (constrained) efficient allocation of consumption and effort we can associate a continuum of combinations of agent's asset holdings and transfer payments. Our specification implies that the planner's value function $V(y_t, U_t, x_t)$ at node h^t corresponds to the expected discounted profits $\Pi_t(\mathcal{W}^*; h^t)$ associated to the optimal contract $\mathcal{W}^*$ at node h^t , when $\mathbf{b}_t^*(h^t) = 0$, and $\boldsymbol{\tau}_t^*(h^t) = \mathbf{c}_t^*(h^t) - y_t$. That is,

$$-V(y_t, U_t, x_t) = -\Pi_t(\mathcal{W}^*; h^t) = \mathbf{E}_t \sum_{\tau=t}^{\infty} \left[\left(\frac{1}{1+r} \right)^{\tau-t} \tau_{\tau}^* \right]$$

is the net present value of transfers agents will receive from period t onwards. According to this interpretation, we can determine a new asset plan $\{\hat{\mathbf{b}}_t(h^t)\}_{t=1}^{\infty}$ as follows

$$\hat{\mathbf{b}}_t(h^t) = -\frac{1}{1+r} \mathbf{E}_t \Pi_{t+1}(\mathcal{W}^*; h^{t+1}) = \mathbf{E}_t \sum_{\tau=t+1}^{\infty} \left[\left(\frac{1}{1+r} \right)^{\tau-t} (\mathbf{c}_{\tau}^*(h^{\tau}) - y_{\tau}) \right]. \quad (46)$$

Considered in cross section average terms, the number $\hat{b}_t$ represents the net position of the economy with respect to the rest of the world, the economywide level of savings in period t . Note, that this definition is consistent with our initial values. That is $0 = \hat{b}_0 = -\frac{1}{1+r} \mathbf{E}_0 \Pi_1(\mathcal{W}^*; y_1)$, where we interpret $\mathbf{E}_0 \Pi_1(\mathcal{W}^*; y_1)$ as the average profit at time 0 (what we fixed to zero by picking U_0 properly). This definition has a clear intuitive meaning from the agent's budget constraint as well. The definition of profits and the agent's budget constraint together with the above definition of implied asset holdings will result in the following transfer scheme

$$\begin{aligned} \hat{\boldsymbol{\tau}}_t(h^t) &= \boldsymbol{\tau}_t^*(h^t) + \mathbf{E}_{t-1} \Pi_t(\mathcal{W}^*; h^t) - \frac{1}{1+r} \mathbf{E}_t \Pi_{t+1}(\mathcal{W}^*; h^{t+1}) \\ &= \mathbf{c}_t^*(h^t) - y_t + (1+r) \hat{\mathbf{b}}_{t-1}(h^{t-1}) - \hat{\mathbf{b}}_t(h^t). \end{aligned}$$

Appendix D: Ex-post Verification

In what follows, we design an artificial testing environment and use the terminology of statistical hypothesis testing in order to valuate how effective is the 'test' we adopted in Section 4.2.4 in distinguishing between models where the first order condition is justified and models where it is not. There, we proposed as ex post deviation measure $R(U_0)$, the percentage improvement in lifetime utility induced by the remaximization, normalized by the percentage of welfare gains with respect to self insurance. The analysis is performed considering two period versions of the model ($T = 2$), because for this case $R(U_0)$ can be computed up to

an arbitrarily small error with very little computational cost (we used a precision of 10^{-10}).

In terms of statistical tests, we will interpret $R(U_0)$ as the relevant *parameter* and consider *the null hypothesis* $R(U_0) \leq 0$. Since we are facing numerical errors, the approximate computation of $R(U_0)$ - denoted by $\hat{R}(U_0)$ - will be interpreted as our *estimate* of the parameter. In order to perform the test we need a test statistic and a rejection rule, i.e. a threshold value for the statistic above which we reject the null. Within this environment, we say that the first order approach is valid if and only if we cannot reject the null. The key reason why in statistical testing one may choose a test statistic different from the estimated parameter itself is because the former has often a known distribution. As we will see below we are not so lucky, hence we will consider $\hat{R}(U_0)$ as our *test statistic* as well. Notice, that for any ‘true’ value for $R(U_0)$, the computed $\hat{R}(U_0)$ is a random variable since it depends both on the specific model (since there are many models delivering the same $R(U_0)$ but possibly leading to a different $\hat{R}(U_0)$) and on the approximation procedure (the specificities of the grid and of the approximation algorithm for example). In what follows we try to infer what a reasonable *critical value* might be by exploring the distribution of $\hat{R}$ across a battery of moral hazard models, some of them where the first order condition approach is justified (i.e. $R(U_0) = 0$), others where $R(U_0) > 0$. Obviously, we cannot consider all possible models, and we are unable to attribute a distribution over the few considered here. On the other hand, we might argue that models very ‘far’ from those where the first order approach is justified (i.e. models with $R(U_0) \gg 0$) are the less relevant to our analysis as it is likely that our test will detect them easily.

Kocherlakota (2004) shows that for the case where agent’s preferences are linear in effort, and effort affects linearly job finding probabilities, the first order approach fails. Notice that this model is concave in both e and c alone hence,⁵⁴ there is a sense in which this framework is not too ‘far’ from the class of models for which we know that the first order approach should work. Therefore, in order to explore the properties of the test statistic $\hat{R}$ when R is close to zero, we construct a class of models where the use of the first order approach cannot be analytically justified by using the following perturbations of the (counter)-example proposed by Kocherlakota. Let $N = 2$, $p(e) = \Pr\{y = y_2^h \mid e\} = e$, and $\frac{c^{1-\sigma}}{1-\sigma} - \eta v(e)$ be the utility function of the agent, with

$$v(e) = \begin{cases} \alpha e^2 & 0 \leq e \leq \underline{e} \\ \psi e + \nu & \text{if } \underline{e} \leq e \leq \bar{e} \\ \frac{\omega}{1-e} + \chi & \bar{e} \leq e \leq 1. \end{cases}$$

Given ψ , $\bar{e}$, and $\underline{e}$ it is possible to uniquely recover the remaining parameters of v maintaining continuity and differentiability everywhere. We fix $\psi = 10$, while the remaining parameters (β , y_2^h , and $y_2^l = y_1$) are set according to Table 1 in Section 4.2. Similarly to the infinite horizon case, we consider two levels of risk

⁵⁴This model fails to be locally jointly concave in (e, b) .

aversion: $\sigma \in \{0.5, 2\}$. Note that for $[\bar{e}, \underline{e}] \subset (0, 1)$, we have $v'(0) = 0$ and $\lim_{e \rightarrow 1} v'(e) = \infty$, therefore the problem always delivers an interior effort solution. Since outside the interval $[\underline{e}, \bar{e}]$ the cost function is strictly concave, we can reduce the degree of linearity of the model by varying, symmetrically, the distance $\bar{e} - \underline{e}$. We chose η in a way that in the limiting case, where $\bar{e} = \underline{e} = 0.5$, the optimal level of effort is 0.5. The other extreme case is given by $[\underline{e}, \bar{e}] = [0, 1]$ which is that studied by Kocherlakota (2004).

Notice that the welfare gain from ex post deviations in the infinite horizon model can be approximated from below by the discounted weighted sum of one period deviations.⁵⁵ In the attempt to obtain a test statistic more comparable to that we would obtain in the infinite horizon case by using two period models, we adopt as relevant parameter $\bar{R}$, the average value of our estimated deviation measure $R(U_0)$ in a sample of U_0 . The choice of the associated test statistic can be problematic if $\hat{R}(U_0)$ varied a lot with U_0 . In our infinite horizon problems however we have not found evidence of large variations. This average measure also assesses the overall validity of the given two period model. The first panel of Table D1 computes $\bar{R}$ as a function of the linearity for the above family of models with negligible numerical errors.⁵⁶ These numbers should hence be seen as values of the true parameter of the model ($R(U_0)$ in terms of the infinite horizon model). The results show a desirable intuitive property of our measure: the closer we are to linearity the more the agent can gain by deviating from the proposed contract hence $\bar{R}$ increases.

Table D1: $\bar{R}$ and $\hat{R}$ for a Family of Linear Models

Degree of Linearity ($\underline{e}; \bar{e}$)	(.1; .9)	(.2; .8)	(.3; .7)	(.4; .6)	(.45; .55)	(.5; .5)
<i>Grid Size: 3×10^{12} (the exact value of $\bar{R}$)</i>						
High Risk Aversion ($\sigma = 2$)	0.1559	0.0501	0	0	0	0
Low Risk Aversion ($\sigma = 1/2$)	1.6328	0.7084	0.2979	0.0787	0.0210	0
<i>Grid Size: 800 ($\hat{R}$)</i>						
High Risk Aversion ($\sigma = 2$)	0.1512	0.0467	0.0065	0.0006	0.0013	0.0015
Low Risk Aversion ($\sigma = 1/2$)	1.6268	0.7073	0.3004	0.0805	0.0199	0.0045

Another interesting observation is that there are no deviations in the four least linear models in the high risk aversion case and in the $\underline{e} = \bar{e} = 0.5$ case in the low risk aversion case. This highlights another desirable property of our verification approach. In these cases, although the agent's problem is not always globally

⁵⁵In fact, in infinite horizon, deviations are likely to accumulate through time as an excessive accumulation of wealth (a deviation in asset holding) is likely to induce the agent to reduce its effort below that recommended. This, in turn, might induce further overaccumulation as low effort reduces the probability of high income levels, and so on.

⁵⁶For Table D1, we obtained the sample of U_0 's by solving the self-insurance model on a grid of first period asset holdings ranging between $-y^l/2$ and y^h .

concave the first-order approach is still valid, implying for example that the agent’s problem is *locally* concave in the neighborhood of the optimal allocation.⁵⁷ This example shows that our numerical approach is not just a substitute for analytical conditions for global concavity, but it can verify the first-order condition approach in a much broader set of models where global concavity cannot be guaranteed or verified analytically.

In AP, we show that when $T = 2$ the procedure should verify the validity of the first order approach in both examples studied in Section 4.2 since the agent’s problem is globally concave in the optimal contract. Hence, we also use the two period versions of the two models studied there as benchmarks for globally concave models (i.e., models where $R(U_0) = 0$).

Table D2: $\hat{R}$ for The Concave Model with Different Grid Sizes

Grid size	200	500	800	1500	2500	5000
High Risk Aversion ($\sigma = 2$)	-0.0191	-0.0015	-0.0013	-0.0002	0.0003	-0.0002
Low Risk Aversion ($\sigma = 1/2$)	0.0030	-0.0001	0.0017	0.0012	0.0008	-0.0001

We will now discuss the results reported in the second panel of Table D1 and Table D2.⁵⁸ In order to infer about the distribution of our test statistic $\hat{R}(U_0)$ in presence of numerical error we computed $\bar{R}$ for all models with coarser grids as well, and denote those numbers by $\hat{R}$. Here we discretized the first period transfer and the second period transfer for low income realization. Then we choose the transfer for high income realization such that this new perturbed transfer scheme gives the same expected discounted income as the original one, given the implemented optimal effort level. In some cases the agent will gain by this perturbed transfer scheme (if the dispersion of second period disposable income shrinks and/or his first period income increases), in other cases it will not be beneficial for him, hence $\hat{R}$ is somehow centered around 0, and it can even take negative values. The key effect of the grid size in this $T = 2$ case is however not on $\hat{R}$, but on the range (dispersion) of re-maximization gains over the sample of U_0 ’s: as we increase the grid size the more concentrated they are around zero.

For the linear models we only report the results of the grid size of 800, because in this case, the implied average distance between grid points is compatible with the that of the infinite horizon model studied in the main text. Table D1 suggests that for the high risk aversion case the critical value should be between 0.0467 and 0.0065. If we choose 0.0467, that minimizes the probability of a false rejection of a model with $\bar{R} = 0$,

⁵⁷Notice that local concavity is a necessary condition for the validity of the first order approach. Our method obviously checks for global deviations as well.

⁵⁸For Table D2, we selected a random sample of 500 life-time utilities representative of the first 500 periods of the infinite horizon models starting from an initial utility level implying zero surplus to the planner. Then we normalized them in order to represent life-time utilities for $T = 2$.

but - by continuity - it is likely to lead to acceptance of some false models. The opposite is true if we choose 0.0165. In this case we maximize the ‘power’ of the test but some true models is likely to be rejected. Any number in between these two values will balance the ‘power’ and ‘significance’ of the test in an intermediate way. In the case of the low risk aversion case, the first panel of Table D1 shows that the null is false in all linear cases, but correct for $\bar{e} = \underline{e} = 0.5$. Therefore, we should set the critical value between 0.0199 and 0.0045. Note that the lower bounds of these intervals are above the measure we obtained for $\hat{R}$ in globally concave models with a similar grid size. Since the first-order condition approach is justified in both cases, this observation highlights that the approximation errors might affect our test statistic to a different extent for different classes of models.

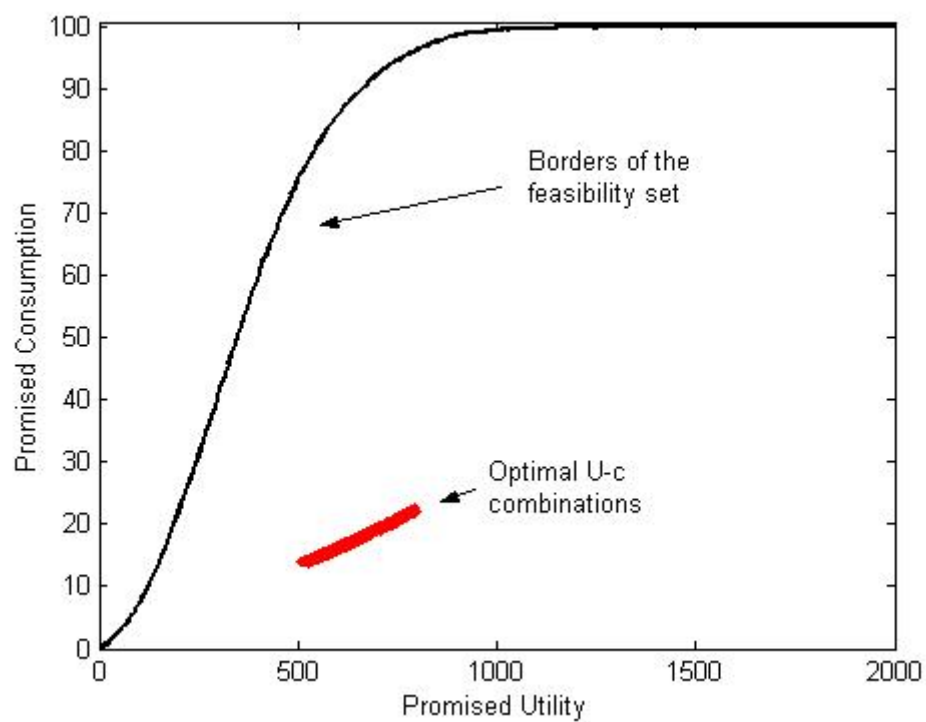


Figure 1A: The domain Restriction and Realized U-c Combinations

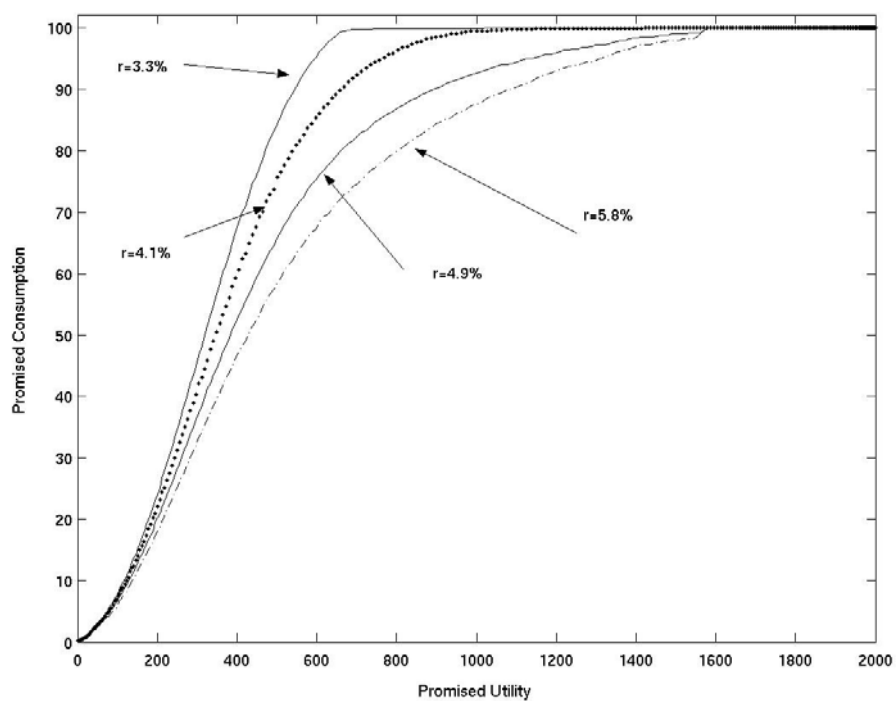


Figure 1B: The domain Restriction for Different Interest Rates

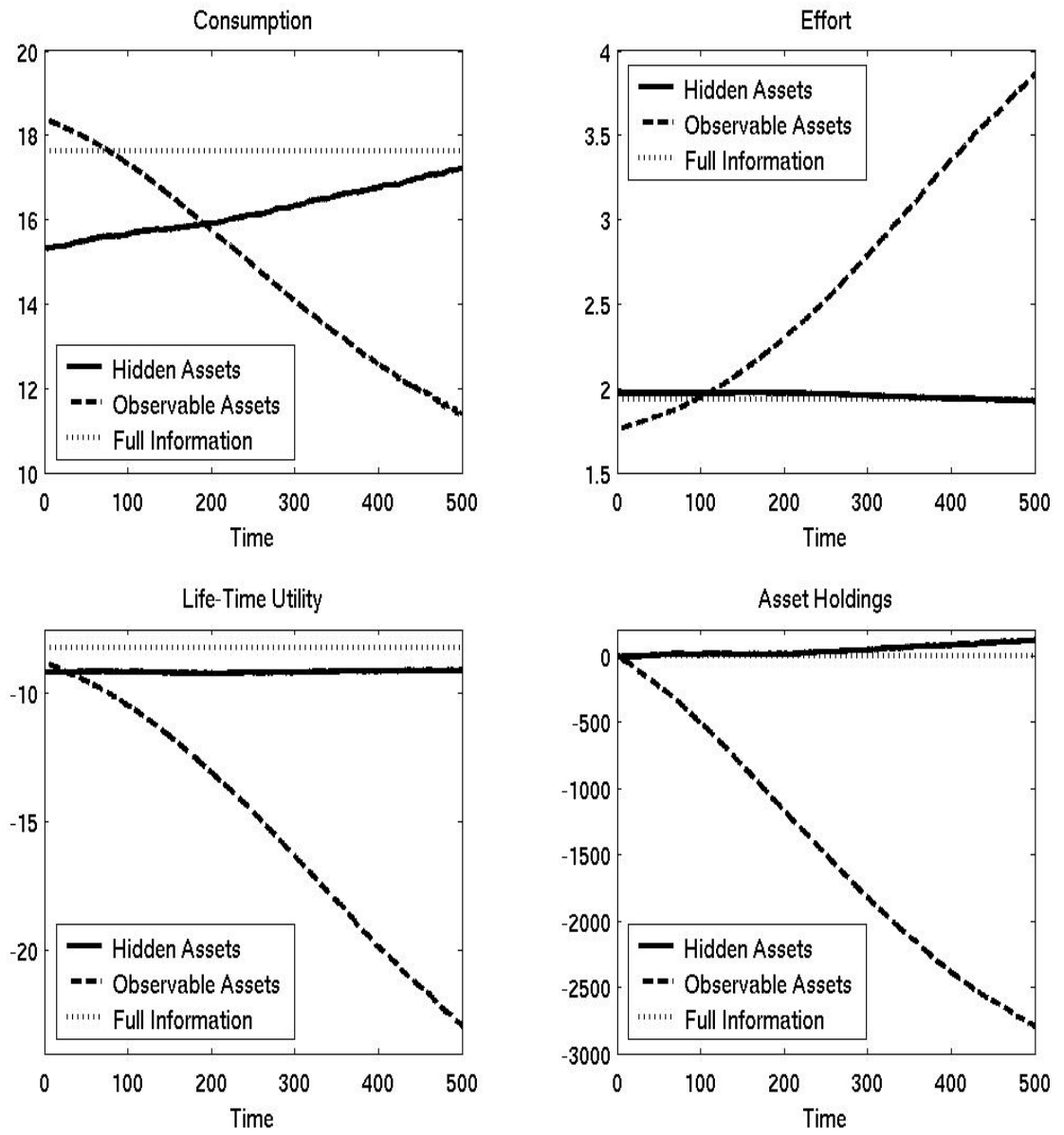


Figure 2: Average path for the first 500 periods
 (U_0 such that the planner has zero expected surplus with the same given initial distribution of output)

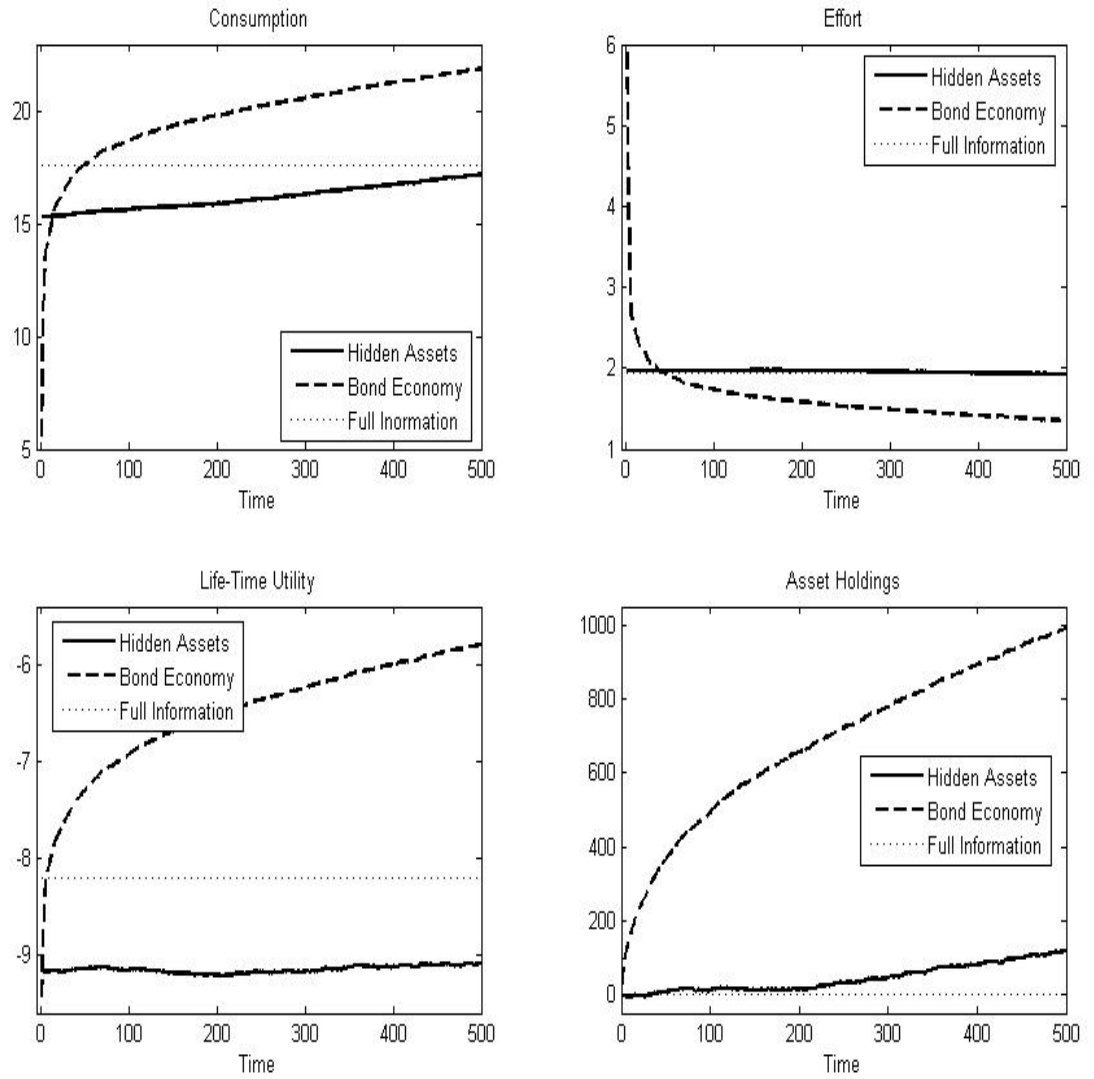


Figure 3: Average path for the first 500 periods
($b_0=0$ for the bond economy and U_0 such that the planner has zero expected surplus with the same given initial distribution of output)

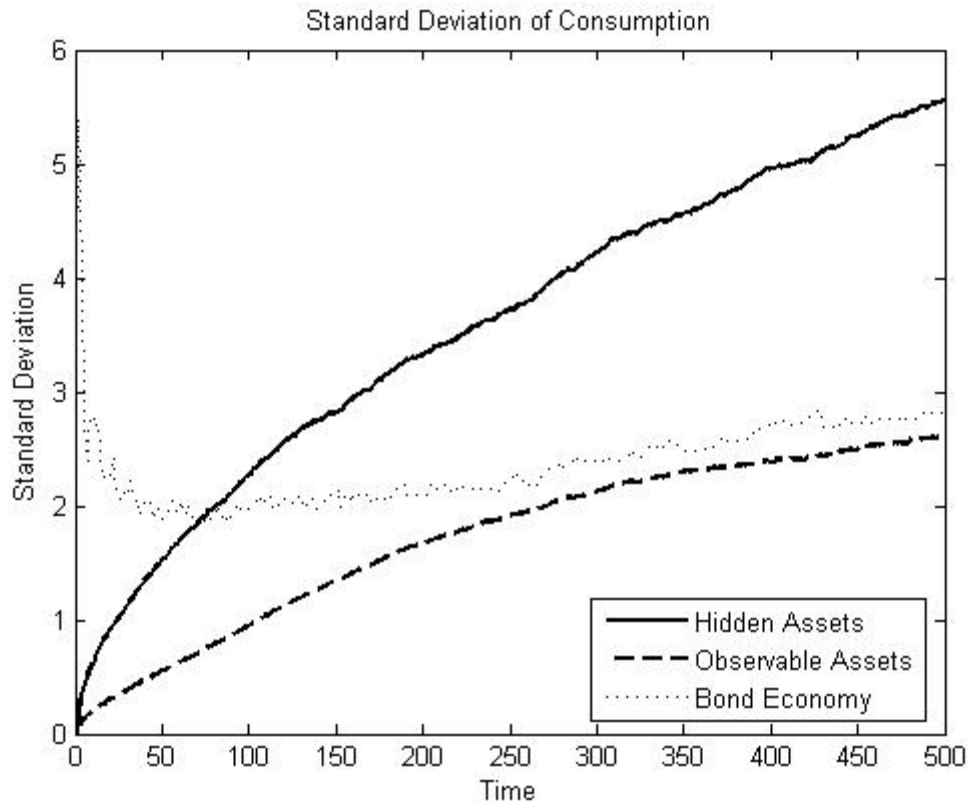


Figure 4A: Average Path for the Standard Deviation of Consumption

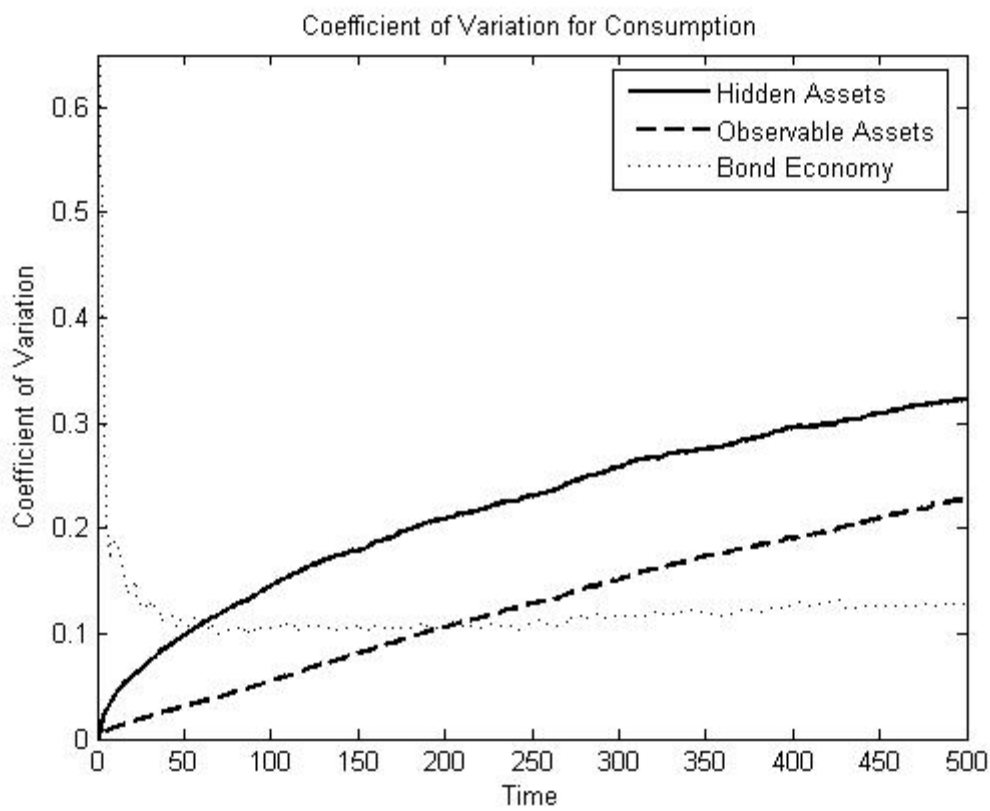


Figure 4B: Average Path for the Coefficient of Variation of Consumption

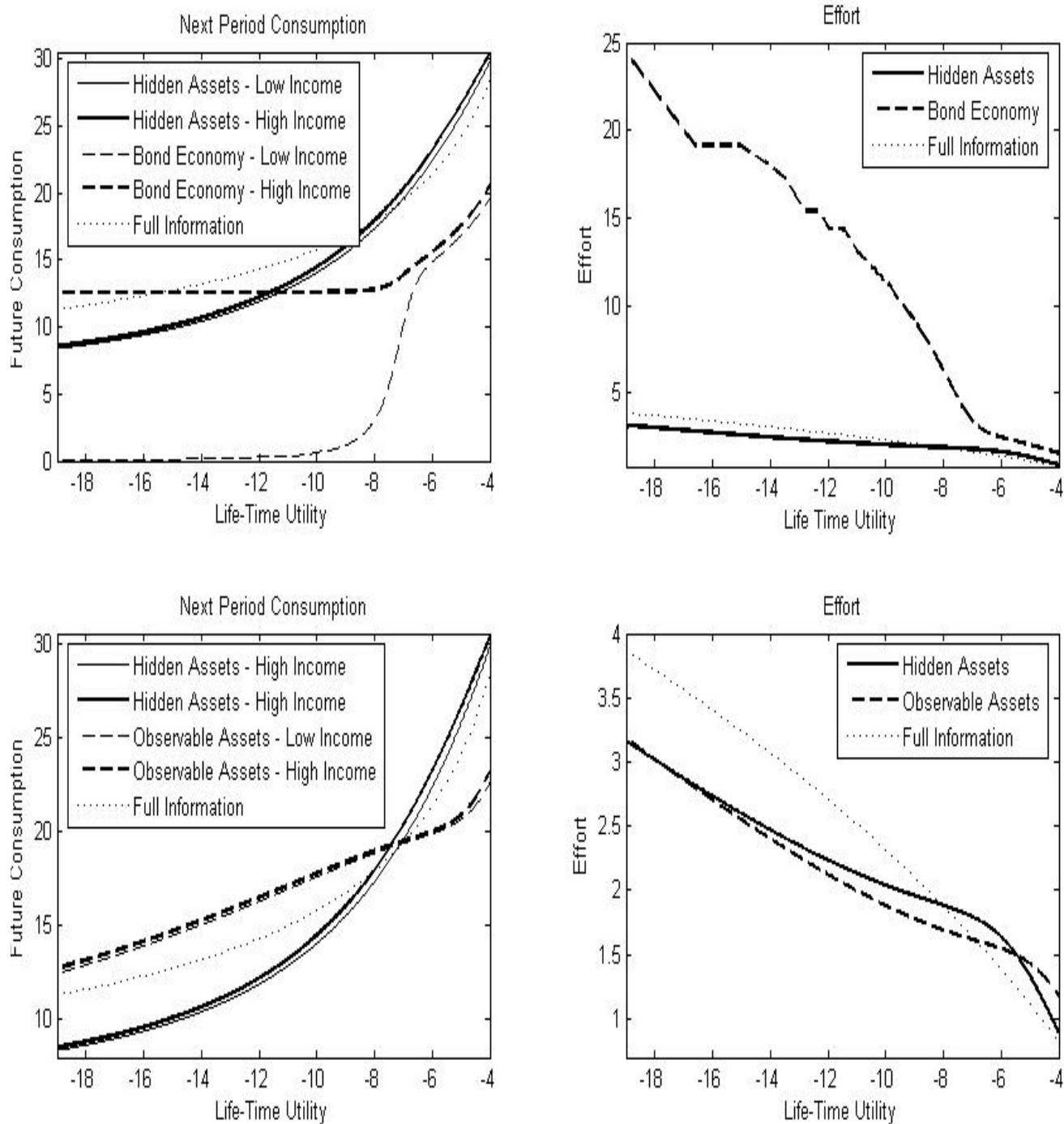


Figure 5: Future Consumption and Effort Policies for the Four Models
 (The life time utility is U_0 for the hidden asset case.)

Table 2: Efficiency gains

	Bond Economy	Full Information	Pure Moral Hazard	Moral Hazard Hidden Assets
<i>High risk aversion ($\sigma=2$)</i>				
<i>$\beta(1+r)=1$ ($r=4.1\%$)</i>				
Agent's life-time utility (%)	100	117.9	112.1	108.3
Consumption increase (%)	0	29.8	19.2	12.9
<i>Welfare gains with higher initial assets $\beta(1+r)=1$ ($r=4.1\%$)</i>				
Agent's life-time utility (%)	100	103.4	102.5	101.8
Consumption increase (%)	0	4.9	3.5	2.6
<i>Low risk aversion ($\sigma=0.5$)</i>				
<i>$\beta(1+r)=1$ ($r=4.1\%$)</i>				
Agent's life-time utility (%)	100	104.3	103.9	103.7
Consumption increase (%)	0	6.6	6.0	5.7
<i>Welfare gains with higher initial assets $\beta(1+r)=1$ ($r=4.1\%$)</i>				
Agent's life-time utility (%)	100	102.1	102.0	102.0
Consumption increase (%)	0	3.7	3.6	3.5
<i>$\beta(1+r)<1$ ($r=3.3\%$)</i>				
Agent's life-time utility (%)	100	110.5	109.2	107.8
Consumption increase (%)	0	16.1	14.1	11.8
<i>$\beta(1+r)>1$ ($r=4.9\%$)</i>				
Agent's life-time utility (%)	100	107.3	106.9	105.3
Consumption increase (%)	0	12.2	11.5	8.8